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High-Performance and Smart Networking Technologies for HPC and AI

A Tutorial Presented at HotI '26 (HotI33)

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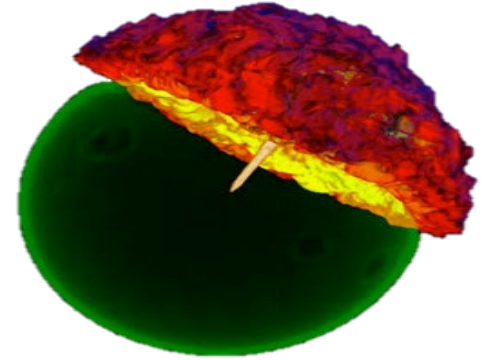
<https://engineering.osu.edu/people/michalowicz.2>

Presentation Overview

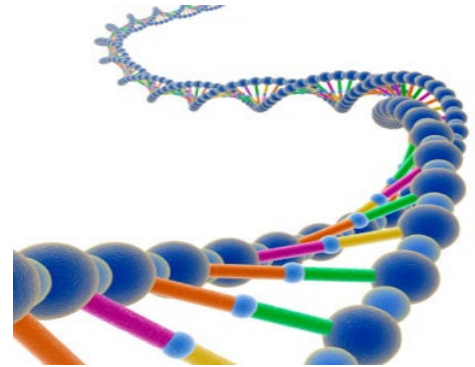
- **Introduction**

- Why High-Performance Networking for HPC and AI?
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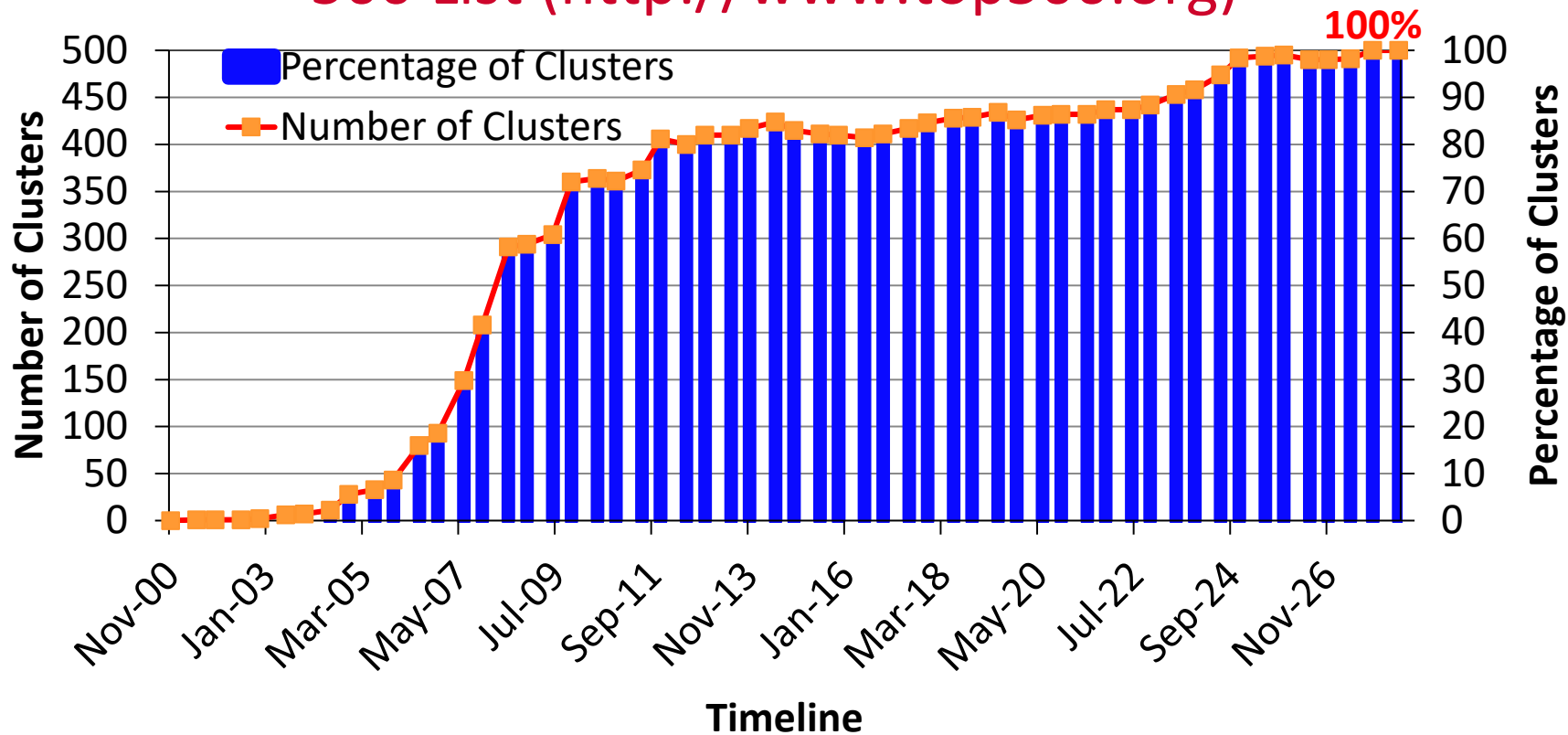
Current and Next Generation Applications and Computing Systems



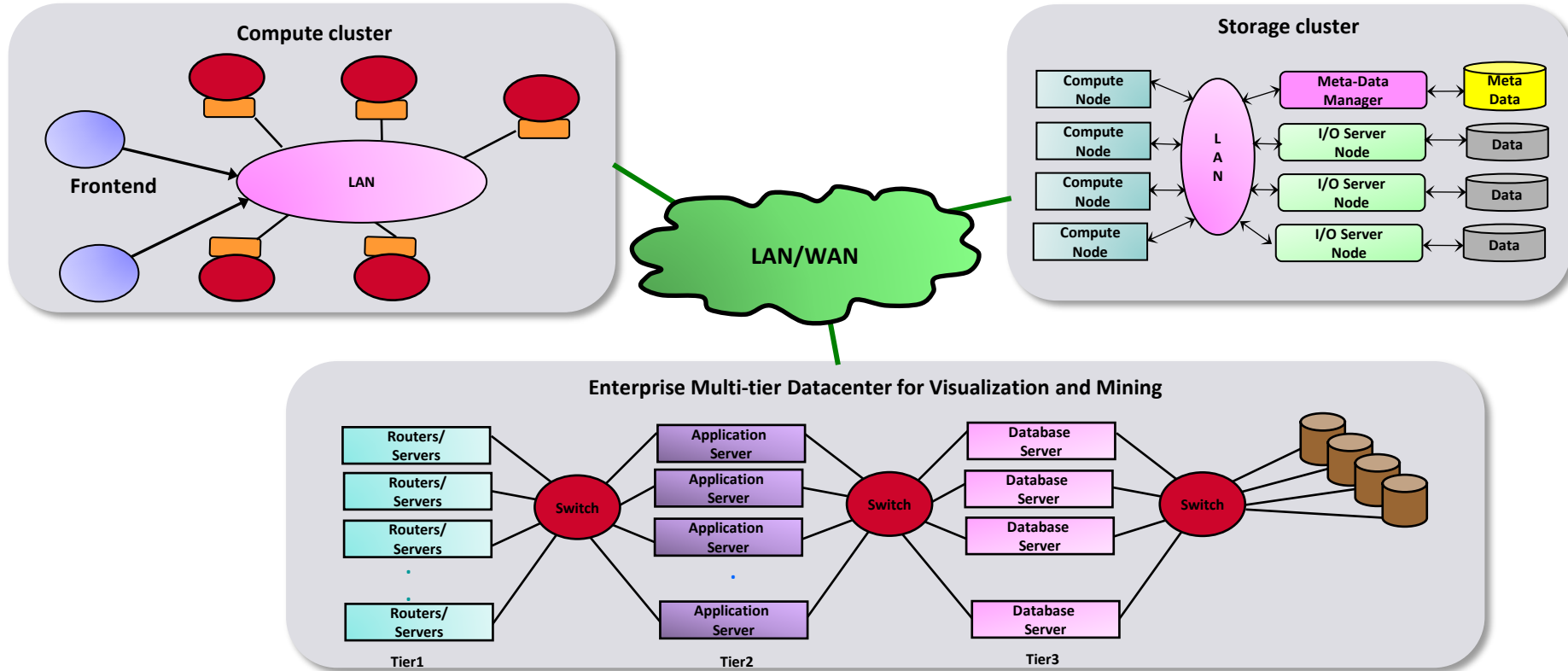
- Growth of High-Performance Computing
 - Growth in processor performance
 - Chip density doubles every 18 months
 - Growth in commodity networking
 - Increase in speed/features + reducing cost
- Clusters: popular choice for HPC
 - Scalability, Modularity and Upgradeability



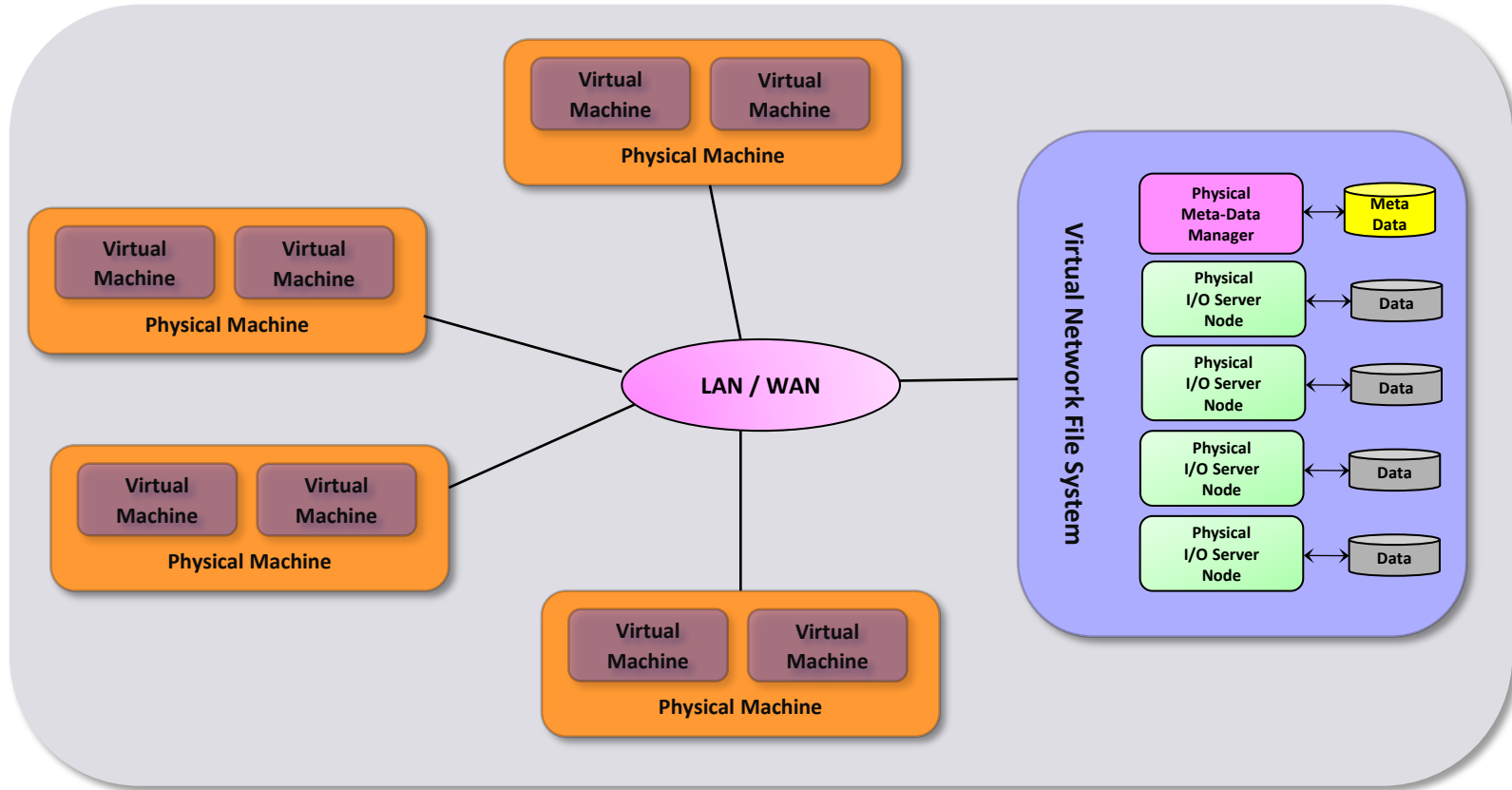
Trends for Commodity Computing Clusters in the Top 500 List (<http://www.top500.org>)

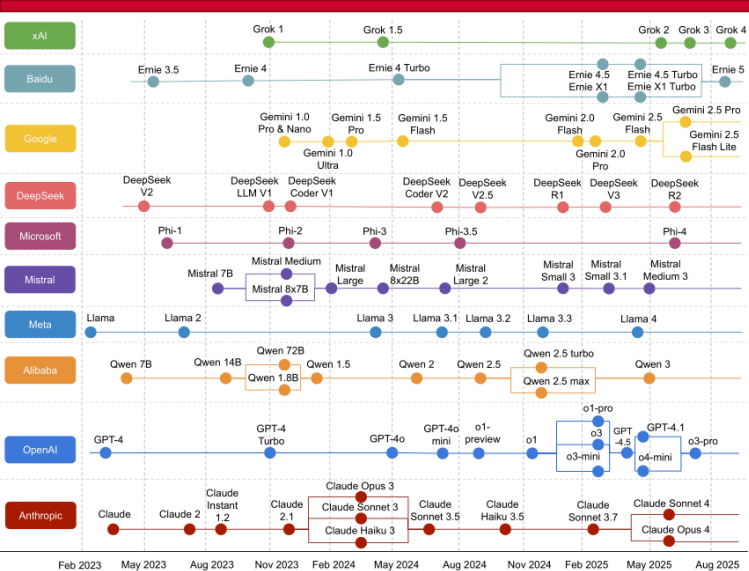


Integrated High-End Computing Environments

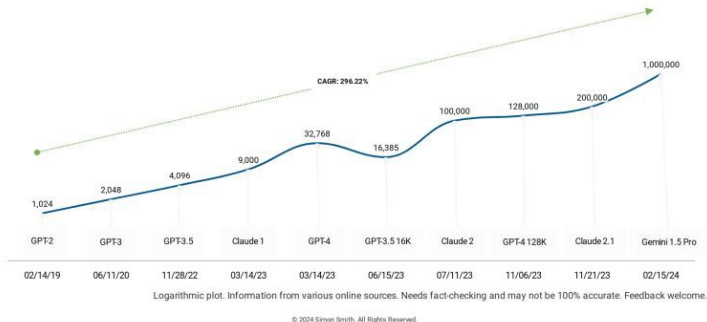


Cloud Computing Environments



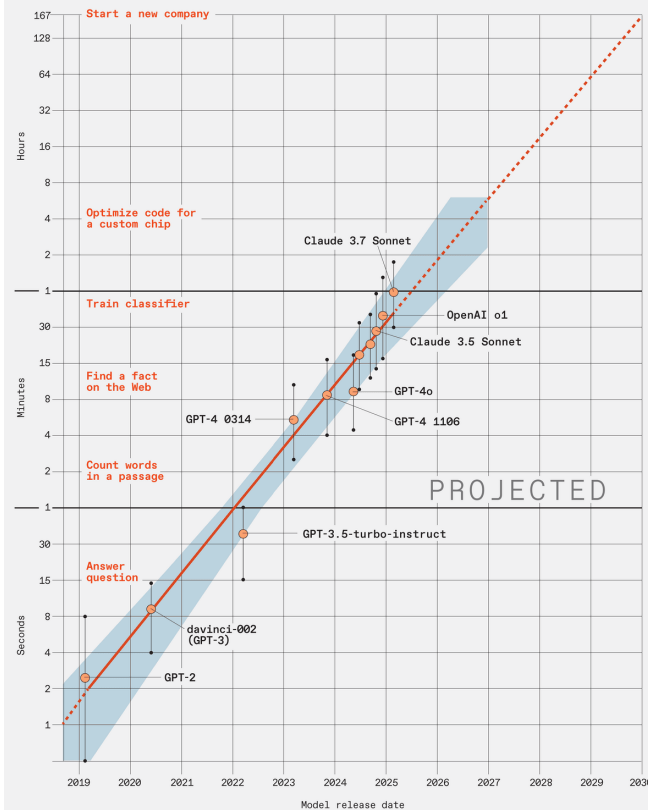


LLM context length increased 997X in five years



Evolution of Language Models

Task Time for a Human That an AI Model Completes With a 50 Percent Success Rate

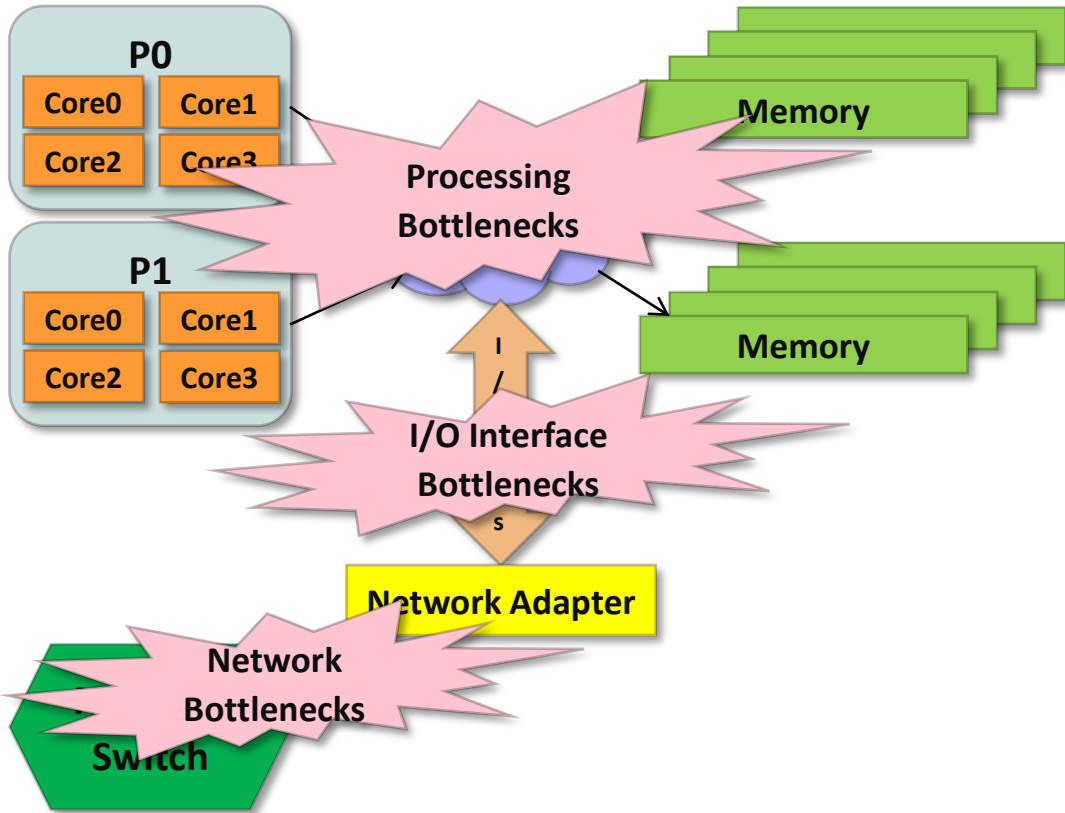


Courtesy: <https://arxiv.org/abs/2507.18479>, https://www.linkedin.com/posts/simonsmith_yesterday-i-speculated-that-you-could-fit-activity-7165048682642432000-Nabx/, <https://spectrum.ieee.org/large-language-model-performance>

Networking and I/O Requirements

- Good System Area Networks with excellent performance (low latency, high bandwidth and low CPU utilization) for inter-processor communication (IPC) and I/O
- Good Storage Area Networks high performance I/O
- Good WAN connectivity in addition to intra-cluster SAN/LAN connectivity
- Quality of Service (QoS) for interactive applications
- RAS (Reliability, Availability, and Serviceability)
- With low cost

Major Components in Computing Systems



- Hardware components
 - Processing cores and memory subsystem
 - I/O bus or links
 - Network adapters/switches
 - GPU, DPU, IPU
- Software components
 - Communication stack
- *Bottlenecks can artificially limit the network performance the user perceives*

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Motivations for High-Performance Networks

- Industry Networking Standards
 - InfiniBand, Ethernet introduced around 2000 to address previous bottlenecks
 - InfiniBand aimed at all three bottlenecks (protocol processing, I/O bus, and network speed)
 - Ultra-Ethernet Consortium Revisits Ethernet inclusion in 2023
- Growing Demand for AI datacenters (“Hyperscaling”)
 - Need for lower latency, higher bandwidth, 100K+ GPU Endpoints and O(millions) CPU endpoints, minimal contention
 - IB, Ethernet, Slingshot, Ultra-Ethernet, GPU-CPU ratios, xPU + Scale-Up networks needed to reduce contention and shift workloads to the network wherever possible
 - Compute moves faster than communication: How to improve the ratio in LLM scaling?

Tackling Communication Bottlenecks with High-Performance Networks

- **Network speed bottlenecks (Improving Bandwidth)**
- Protocol processing bottlenecks (Improving Latency)
- I/O interface bottlenecks (Improving Bandwidth and Latency)

Network Bottleneck Alleviation: Case Study With InfiniBand (“Infinite Bandwidth”) and High-speed Ethernet

- Bit serial differential signaling
 - Independent pairs of wires to transmit independent data (called a lane)
 - Scalable to any number of lanes
 - Easy to increase clock speed of lanes (since each lane consists only of a pair of wires)
- Theoretically, no perceived limit on the bandwidth
 - Scaling up clock speed → correlation to lower latency



Network Speed Acceleration since 2001

**800
times in
the last
25
years!!**

InfiniBand (2001 -)	2 Gbit/sec (1X SDR)
10-Gigabit Ethernet (2001 -)	10 Gbit/sec
InfiniBand (2003 -)	8 Gbit/sec (4X SDR)
InfiniBand (2005 -)	16 Gbit/sec (4X DDR)
	24 Gbit/sec (12X SDR)
InfiniBand (2007 -)	32 Gbit/sec (4X QDR)
40-Gigabit Ethernet (2010 -)	40 Gbit/sec
InfiniBand (2011 -)	54.6 Gbit/sec (4X FDR)
InfiniBand (2012 -)	2 x 54.6 Gbit/sec (4X Dual-FDR)
25-/50-Gigabit Ethernet (2014 -)	25/50 Gbit/sec
100-Gigabit Ethernet (2015 -)	100 Gbit/sec
Omni-Path (2015 -)	100 Gbit/sec
InfiniBand (2015 -)	100 Gbit/sec (4X EDR)
InfiniBand (2017 -)	200 Gbit/sec (4X HDR)
Slingshot10/11 (HPE Slingshot 200) (2021 -)	200 Gbit/sec
Omni-Path-Express (2021 -)	100 Gbit/sec
Google Aquila (2021 -)	100 Gbit/sec
800-Gigabit Ethernet	800 Gbit/sec
InfiniBand (2022 -)	400 Gbit/sec (4X NDR)
HPE Slingshot 400 (2025 -)	400 Gbit/sec
Omni-Path-Express (2024 -)	400 Gbit/sec (CN5000)
InfiniBand (2025 -)	800 Gbit/sec (4X XDR)
Celestica DS6000/6001 (2025 -)	1.6 Tbit/sec

Tackling Communication Bottlenecks with High-Performance Networks

- Network speed bottlenecks (Improving Bandwidth)
- **Protocol processing bottlenecks (Improving Latency)**
- **I/O interface bottlenecks**

Capabilities of High-Performance Networks

- Intelligent Network Interface Cards → Smart Network Cards (“SmartNIC”)
 - Datacenter Processing Units/Infrastructure Processing Units (DPU/IPU)
- Support entire protocol processing completely in hardware (hardware protocol offload engines)
- Provide a rich communication interface to applications
 - *User-level communication capability*
 - Gets rid of intermediate data buffering requirements
- No software signaling between communication layers
 - All layers are implemented on a *dedicated* hardware unit, and not on a *shared* host CPU

Upcoming and Current I/O Interface Architectures

- Compute Express Link (CXL) – cache-coherent for cross-device (storage, xPU, accelerator) communication
 - <https://computeexpresslink.org/>
- NVLink/NVLink-Fusion/NVLink-SHARP – NVIDIA Scale-Up for CPUs/GPUs; avoids contention through PCIe Bus in CPU-GPU transfer
 - <http://nvidia.com/object/nvlink.html>
- AMD Infinity Fabric – AMD Scale-Up for GPU/APU
 - <https://amd.com/en/blogs/2025/engineering-the-future-of-ai.html>
- Ultra-Accelerator Link Consortium – Accelerator Agnostic Scale-Up interconnect
 - <https://ualinkconsortium.org>
- Ultra Ethernet Consortium – Also building Scale-Up Ethernet solutions within a server
 - <https://ultraethernet.org/>

Ultra Ethernet
Consortium

CXL Compute Express Link®

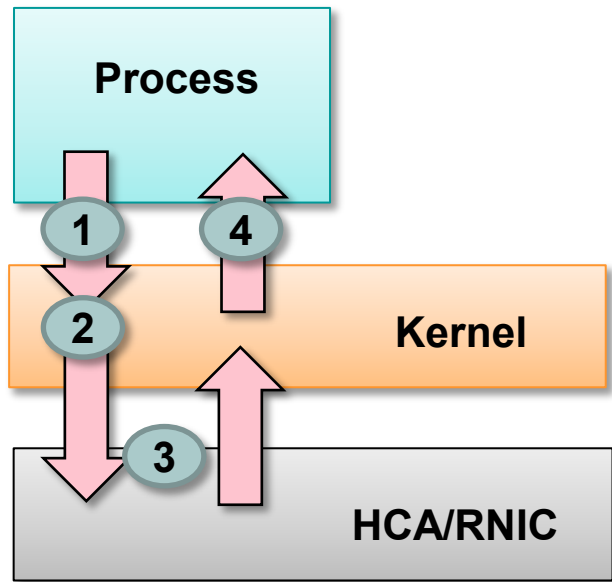


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Memory Registration

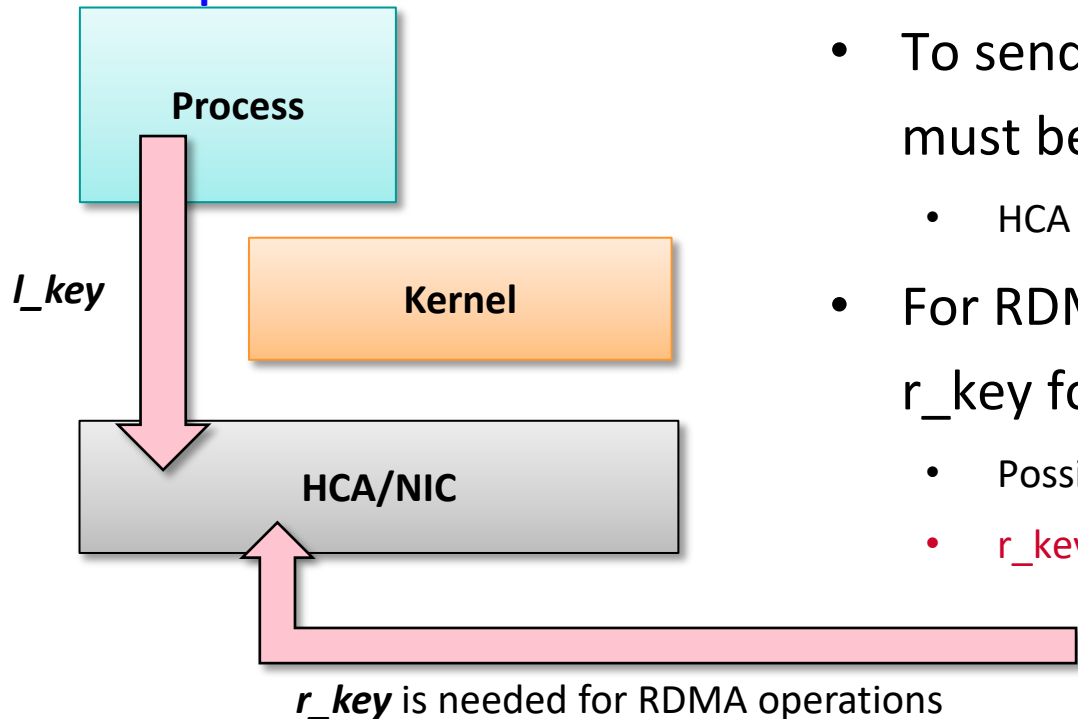
**Before we do any communication:
All memory used for communication must
be registered**



1. Registration Request
 - Send virtual address and length
2. Kernel handles virtual->physical mapping and pins region into physical memory
 - Process cannot map memory that it does not own (security !)
3. HCA caches the virtual to physical mapping and issues a handle
 - Includes an *l_key* and *r_key*
4. Handle is returned to application
 - Incurs initial cost, but allows for kernel bypass to:
 - Avoid context switching (expensive) – reduces latency
 - Perform Remote Direct Memory Access (RDMA) – one-sided communication (reduces latency and improves bandwidth)

Memory Protection

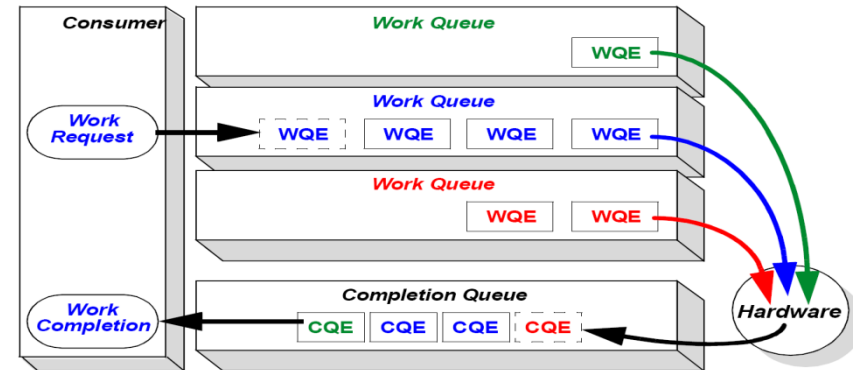
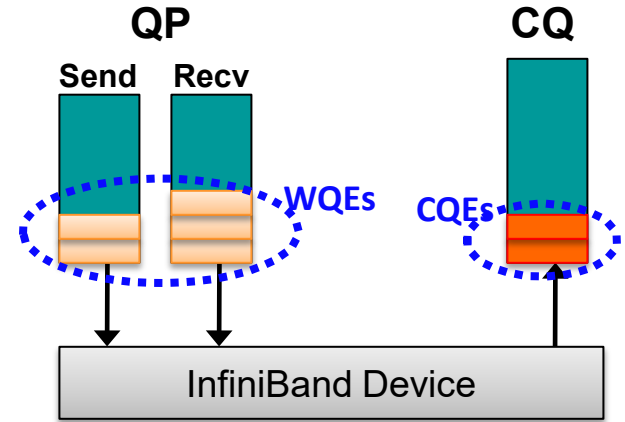
For security, keys are required for all operations that touch buffers



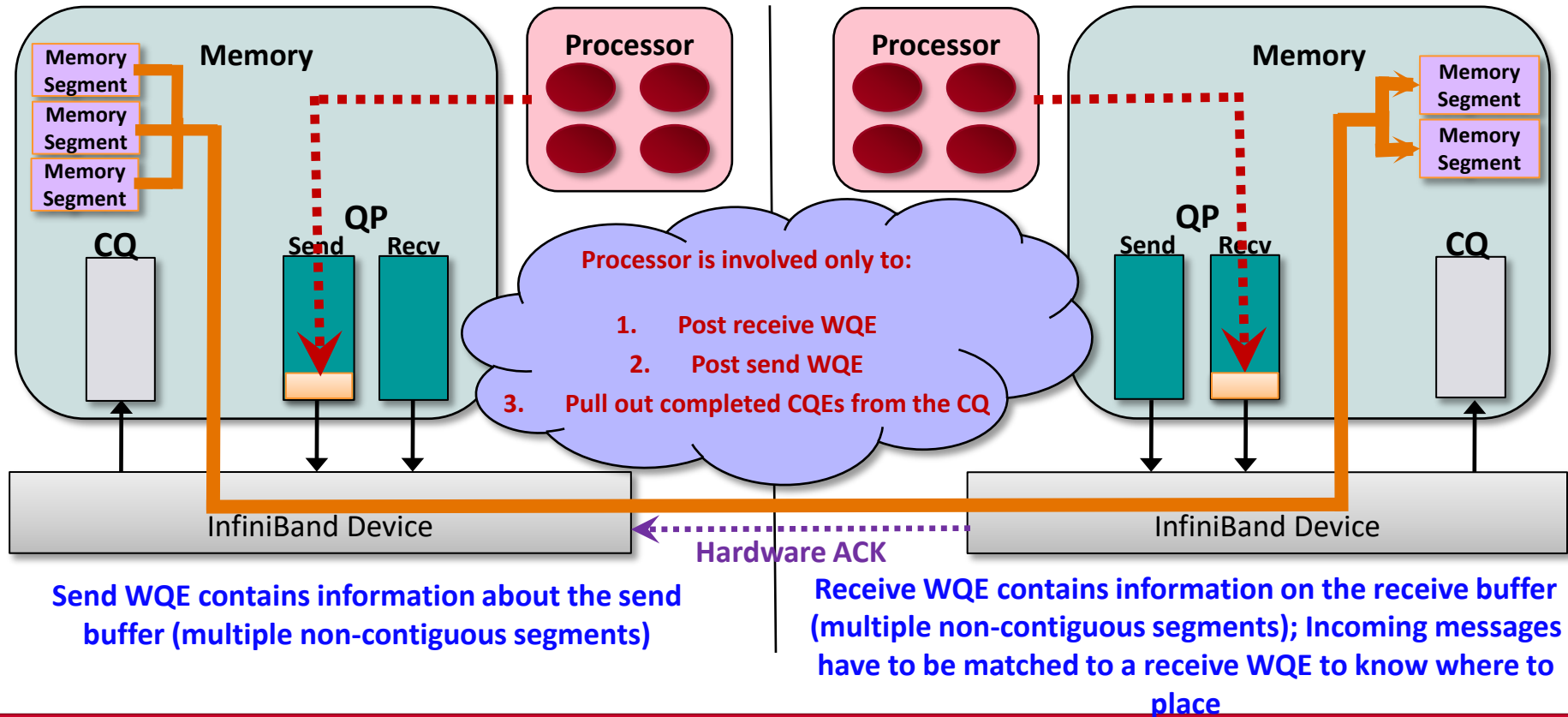
- To send or receive data the *l_key* must be provided to the HCA
 - HCA verifies access to local memory
- For RDMA, initiator must have the *r_key* for the remote virtual address
 - Possibly exchanged with a send/recv
 - *r_key* is not encrypted in IB

Queue Pair Model

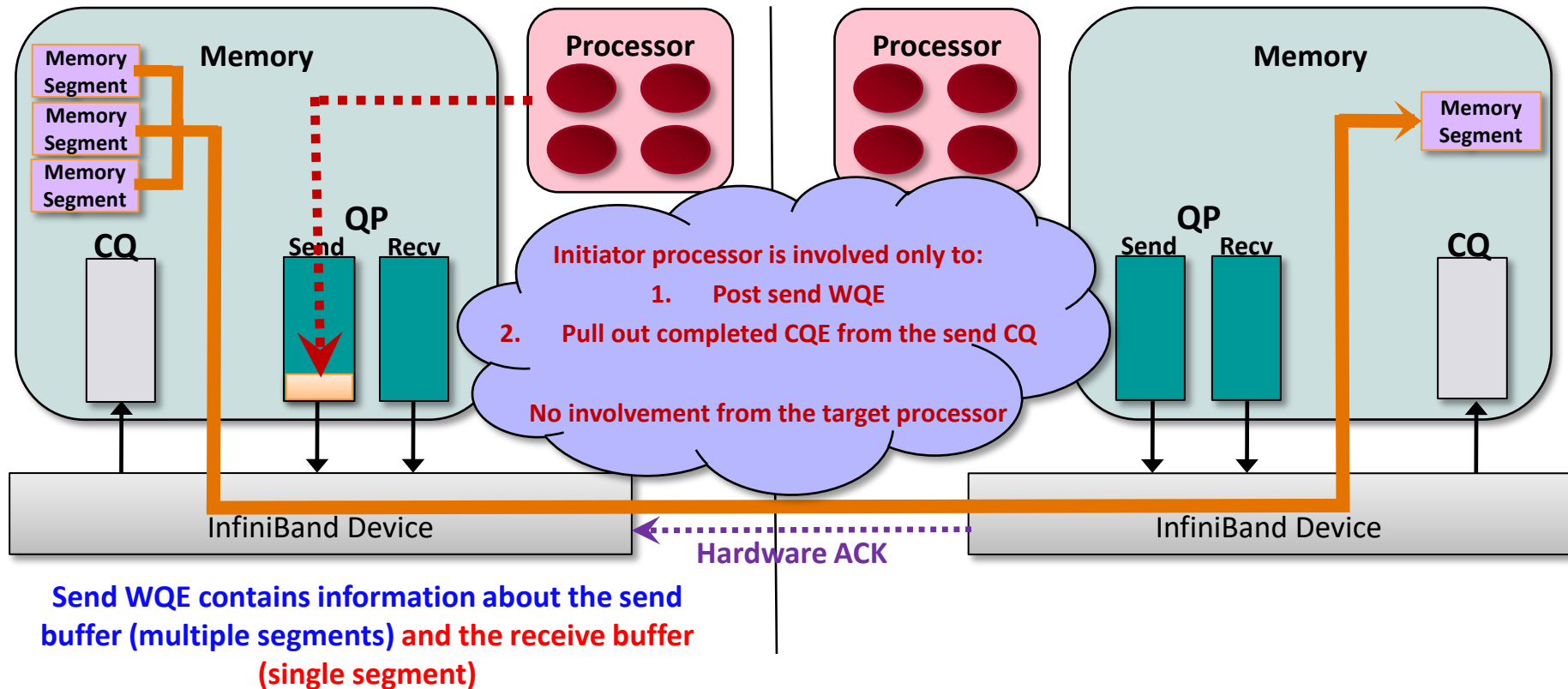
- Each QP has two queues
 - Send Queue (SQ)
 - Receive Queue (RQ)
 - Work requests are queued to the QP (WQEs: “Wookies”)
- QP to be linked to a Completion Queue (CQ)
 - Gives notification of operation completion from QPs
 - Completed WQEs are placed in the CQ with additional information (CQEs: “Cookies”)



Communication in the Channel Semantics (Send/Receive Model)



Communication in the Memory Semantics (RDMA Model)



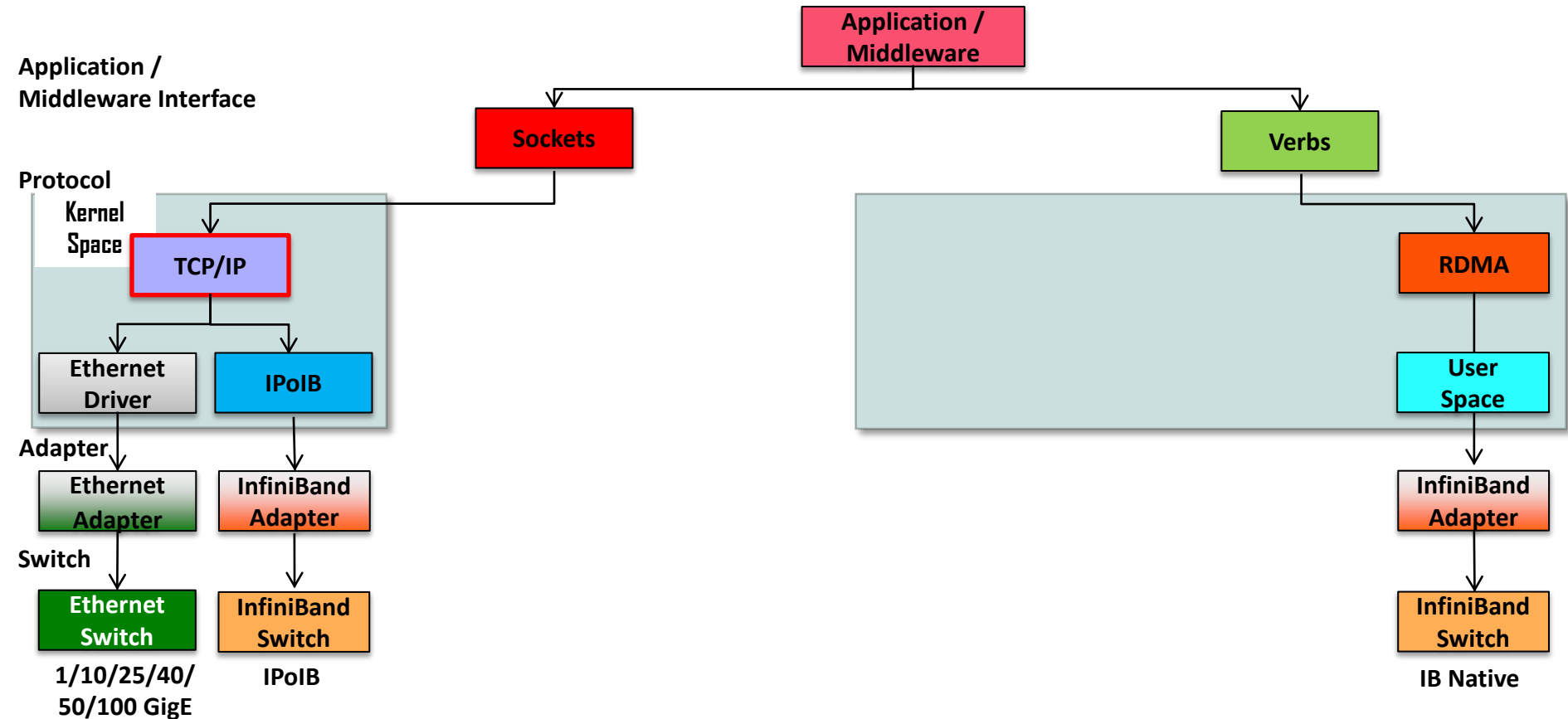
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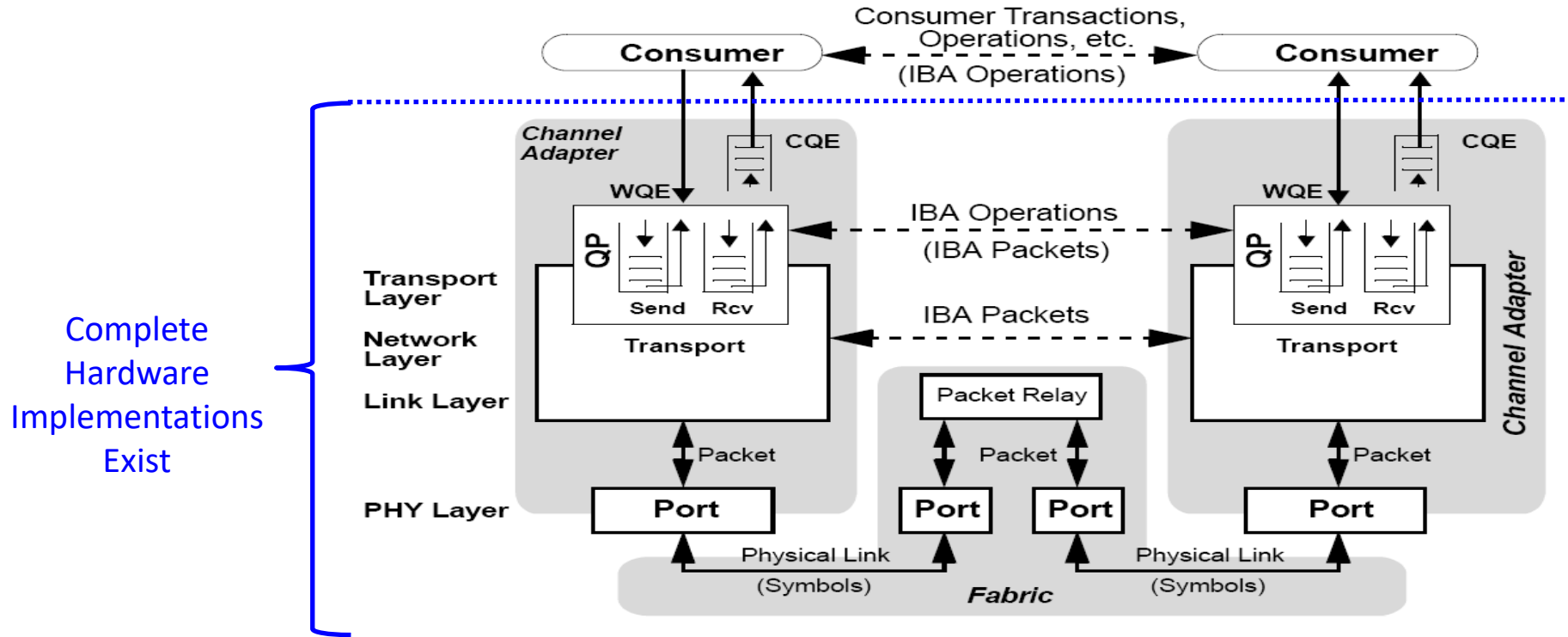
IB, HSE and their Convergence

- **InfiniBand**
 - **Breakdown with respect to TCP/IP**
 - **Hardware Offload Protocol**
- **InfiniBand/Ethernet Convergence Technologies**
 - (InfiniBand) RDMA over Converged (Enhanced) Ethernet (RoCE)

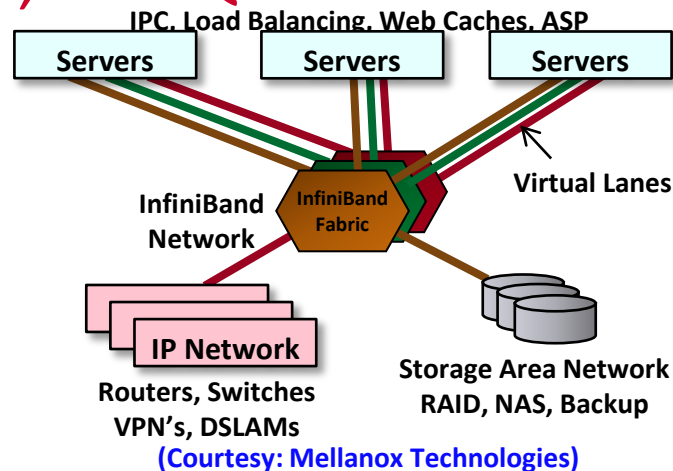
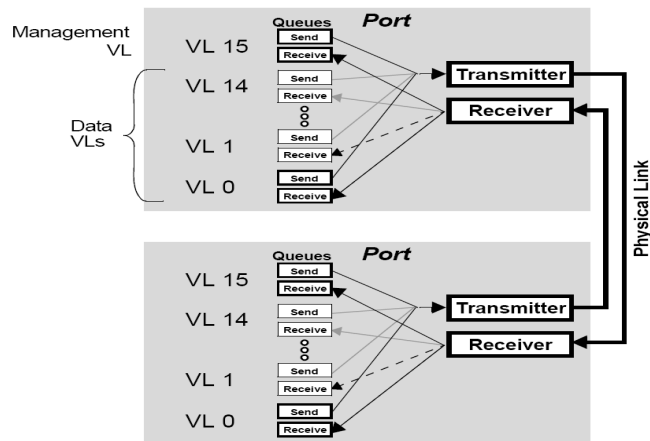
TCP/IP, IPoIB and Native IB Verbs



Hardware Protocol Offload



Virtual Lanes, Service Levels, and QoS



Traffic Segregation

- Virtual Lanes (VL)
 - Multiple (between 2 and 16) virtual links within same physical link
 - 0 – default data VL; 15 – VL for management traffic
 - Separate buffers and flow control
 - Avoids Head-of-Line Blocking
- Service Level (SL):
 - Packets may operate at one of 16, user defined SLs
- SL to VL mapping:
 - SL determines which VL on the next link is to be used
 - Each port (switches, routers, end nodes) has a SL to VL mapping table configured by the subnet management
- Partitions:
 - Fabric administration (through Subnet Manager) may assign specific SLs to different partitions to isolate traffic flows

IB Transport Types and Associated Trade-offs

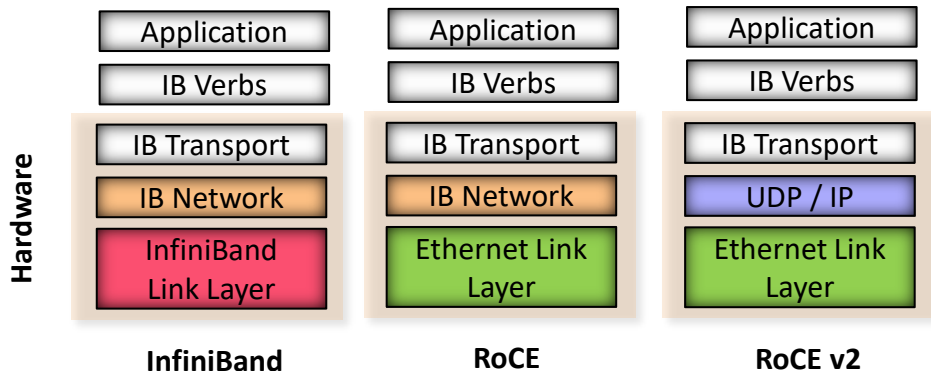
Attribute		Reliable Connection (RC)	Reliable Datagram (RD)	Dynamic Connected (DC/DCT)	eXtended Reliable Connection (XRC)	Unreliable Connection (UC)	Unreliable Datagram (UD)	Raw Datagram
Scalability (M processes, N nodes)		M ² N QPs per HCA	M QPs per HCA	M QPs per HCA	MN QPs per HCA	M ² N QPs per HCA	M QPs per HCA	1 QP per HCA
Reliability	Corrupt data detected	Yes						
	Data Delivery Guarantee	Data delivered exactly once				No guarantees		
	Data Order Guarantees	Per connection	One source to multiple destinations	Per connection	Per connection	Unordered, duplicate data detected	No	No
	Data Loss Detected	Yes					No	No
	Error Recovery	Errors (retransmissions, alternate path, etc.) handled by transport layer. Client only involved in handling fatal errors (links broken, protection violation, etc.)				Packets with errors and sequence errors are reported to responder	None	None

IB, HSE and their Convergence

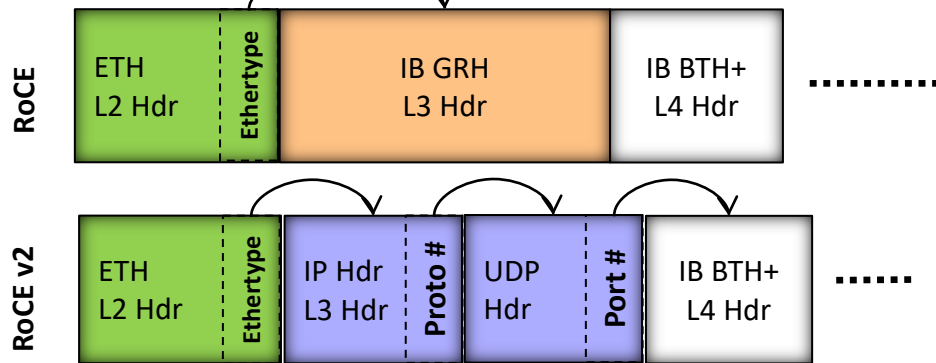
- InfiniBand
 - Architecture and Basic Hardware Components
 - Novel Features
- **InfiniBand/Ethernet Convergence Technologies**
 - **(InfiniBand) RDMA over Converged (Enhanced) Ethernet (RoCE)**

RDMA over Converged Enhanced Ethernet (RoCE)

Network Stack Comparison



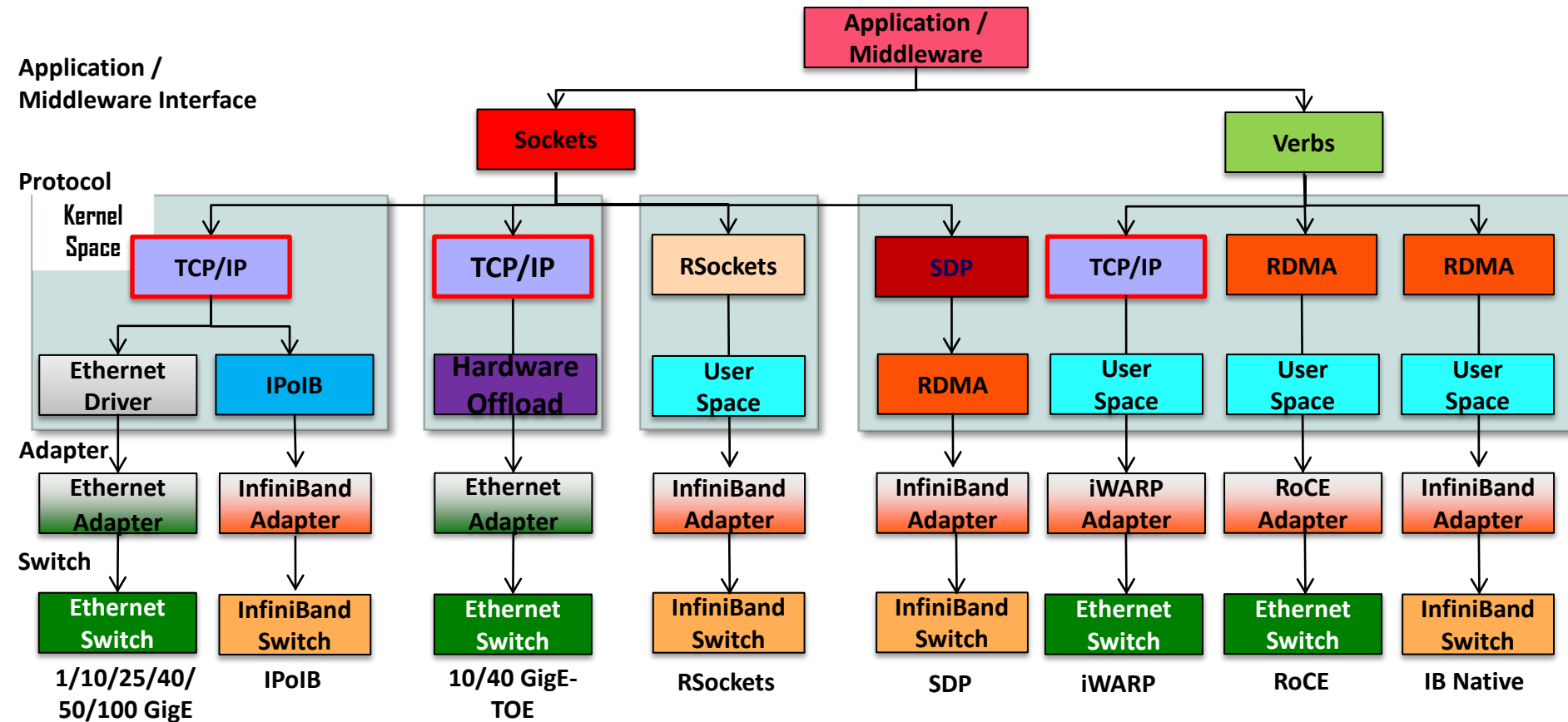
Packet Header Comparison



Courtesy: OFED, Mellanox

- Takes advantage of IB and Ethernet
 - Software written with IB-Verbs
 - Link layer is Converged (Enhanced) Ethernet (CE)
 - 100Gb/s support from latest EDR and ConnectX-3 Pro adapters
- Pros: IB Vs RoCE
 - Works natively in Ethernet environments
 - Entire Ethernet management ecosystem is available
 - Has all the benefits of IB verbs
 - Link layer is very similar to the link layer of native IB, so there are no missing features
- RoCE v2: Additional Benefits over RoCE
 - Traditional Network Management Tools Apply
 - ACLs (Metering, Accounting, Firewalling)
 - GMP Snooping for Optimized Multicast
 - Network Monitoring Tools

RDMA over Converged Ethernet (RoCE)



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Getting Set-up for the Hands-on Exercise 1 (IB Performance)

- You will run the experiments on the OSU RI2 cluster

- Please use the account name and password from <http://go.osu.edu/ibtutorial>**

- Open your favorite Terminal

\$ ssh ri2tut01@ri2.cse.ohio-state.edu

Enter Password:

- Make sure to `cd` to your home directory before cloning the repository

- Clone from <https://github.com/OSU-Nowlab/HPN-Tutorial.git>
- Verify that the following files are present:

```
[michalowicz.2@ri2 HPN-Tutorial]$ ls -l
total 56
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 481 Jul 24 18:36 README.TXT
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 1801 Jul 27 13:02 run_dpu-rome.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 1379 Jul 27 13:01 run_dpu-thor.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 624 Jul 24 18:36 run_fabtest_bw.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 640 Jul 24 18:36 run_fabtest_lat.sh
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 1356 Jul 27 13:02 run_gpu.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 673 Jul 24 18:36 run_inline_ib.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 668 Jul 24 18:36 run_inline_mpi.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 683 Jul 24 18:36 run_mtu.sh
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 433 Jul 24 18:36 run_omb_ofi.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 444 Jul 24 18:36 run_omb.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 498 Jul 24 18:36 run_perftest_bw.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 501 Jul 24 18:36 run_perftest_lat.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 866 Jul 24 18:36 run_rc_ud.sh
```

Step 2: Benchmarking InfiniBand Latency

```
$ srun -N2 --reservation=ibtutorial run_perftest_lat.sh
```

```
Executing '/usr/bin/ib_send_lat -d mlx5_0 -a' on the server
```

```
Executing '/usr/bin/ib_send_lat -d mlx5_0 -a gpu08' on the client
```

```
. . . .
```

#bytes percentile[usec]	#iterations	t_min[usec]	t_max[usec]	t_typical[usec]	t_avg[usec]	t_stdev[usec]	99% percentile[usec]	99.9%
2	1000	0.94	3.00	0.98	0.98	0.04	1.08	3.00
4	1000	0.94	3.36	0.96	0.98	0.04	1.04	3.36
8	1000	0.95	2.96	0.99	0.99	0.04	1.09	2.96
16	1000	0.95	2.83	0.97	0.98	0.04	1.01	2.83
32	1000	1.02	2.60	1.04	1.05	0.00	1.14	2.60
64	1000	1.00	2.15	1.03	1.04	0.00	1.07	2.15
128	1000	1.07	3.09	1.12	1.12	0.03	1.14	3.09
256	1000	1.45	3.21	1.53	1.54	0.03	1.67	3.21
512	1000	1.51	4.22	1.54	1.56	0.03	1.69	4.22
1024	1000	1.64	3.45	1.67	1.69	0.03	1.79	3.45...

Benchmarking InfiniBand Latency

#bytes	ib_send_lat	ib_write_lat	ib_read_lat
2	1.01	0.94	1.93
4	1.01	0.94	1.96
8	1.01	0.95	1.97
16	1.02	0.95	1.98
32	1.02	1.02	1.98
64	1.12	1	2
128	1.16	1.07	2.05
256	1.60	1.45	2.1
512	1.64	1.51	2.14
1024	1.78	1.64	2.4
2048	2.01	1.88	2.64
4096	2.48	2.34	3.12
8192	3.16	3.02	3.79
16384	4.34	4.19	5.03
32768	6.41	6.25	7.05

write is faster

Latency Increases with Message Size

Step 3: Benchmarking InfiniBand Bandwidth

```
$ srun -N2 --reservation=ibtutorial run_perftest_bw.sh
```

```
Executing '/usr/bin/ib_send_bw -d mlx5_0 -a' on the server
```

```
Executing '/usr/bin/ib_send_bw -d mlx5_0 -a gpu08' on the client
```

```
. . . . .
```

#bytes	#iterations	BW peak[MB/sec]	BW average[MB/sec]	MsgRate[Mpps]
2	1000	12.01	11.85	6.210454
4	1000	22.17	17.74	4.650813
8	1000	45.55	35.99	4.716851
16	1000	90.87	71.80	4.705226
32	1000	179.96	143.74	4.710197
64	1000	362.58	286.97	4.701794
128	1000	726.98	573.51	4.698231
256	1000	1457.57	1111.64	4.553276
512	1000	2872.28	2281.59	4.672705
1024	1000	5488.89	4384.14	4.489363

Benchmarking InfiniBand Bandwidth

#Bytes	ib_send_bw	ib_write_bw	ib_read_bw
2	11.59	11.68	12.01
4	15.84	23.78	22.17
8	45.1	47.07	45.55
16	90.2	93.66	90.87
32	176.06	188.28	179.96
64	351.28	377.54	362.58
128	719.83	751.2	726.98
256	1385.21	1498.58	1457.57
512	2694	2974.33	2872.28
1024	5540.84	5888.88	5488.89
2048	9664.95	9868.51	6823.24
4096	10393.64	10825.72	6304.71
8192	11140.84	11180.81	9600.63
16384	11197.53	11363.68	9847.76
32768	11274.91	11295.33	10802.41

Bandwidth Increases
until Link is Saturated



Presentation Overview

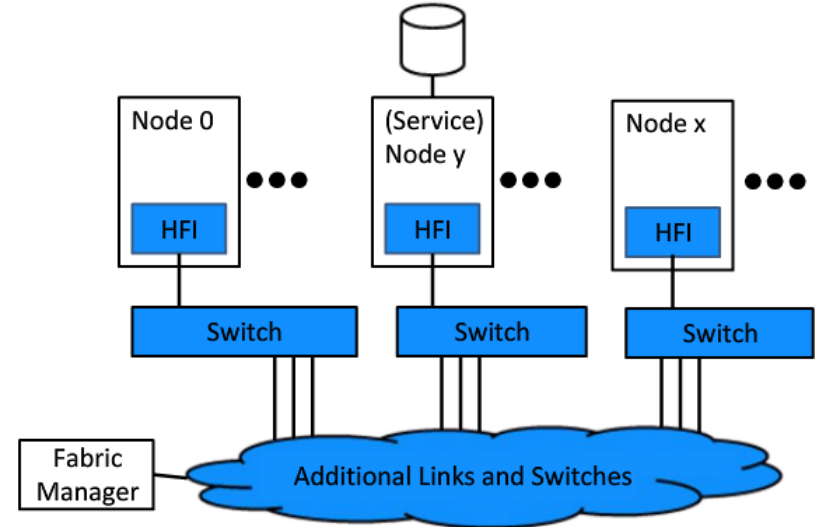
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A Brief History of Omni-Path

- Pathscale (2003 – 2006) came up with initial version of IB-based product
- QLogic enhanced the product with the PSM software interface
- The IB product of QLogic was acquired by Intel
- Intel enhanced the QLogic IB product to create the Omni-Path product
- Sustained for several years
- Intel declared the product being dead
- Currently pursued by Cornelis Networks (Omni-Path Express)
 - CN5000 runs 400Gbps, with current installations in large clusters such as Texas Advanced Computing Center's "Stampede3" Deployment

Omni-Path Fabric Overview

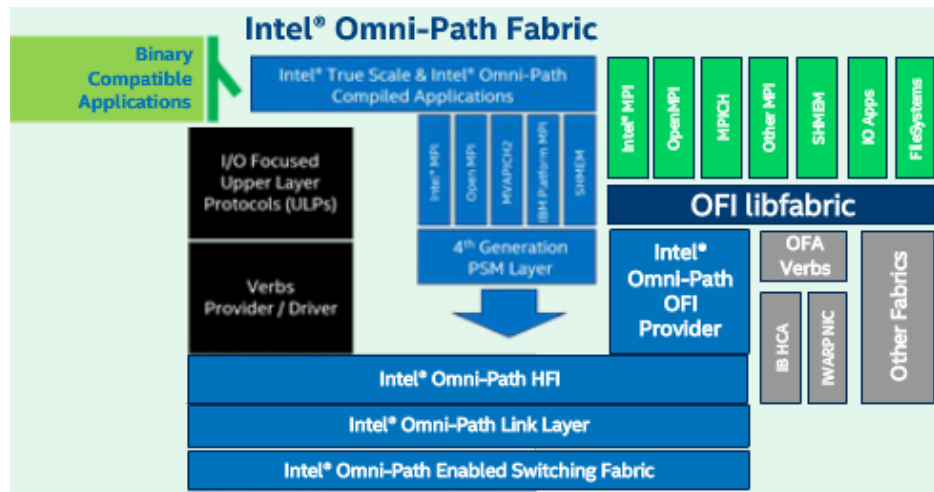
- **Layer 1.5: Link Transfer Protocol**
 - Features
 - Traffic Flow Optimization
 - Packet Integrity Protection
 - Dynamic Lane Switching
 - Error detection/replay occurs in Link Transfer Packet units
 - 1 Flit = 65b; LTP = 1056b = 16 flits + 14b CRC + 2b Credit
 - LTPs implicitly acknowledged
 - Retransmit request via NULL LTP; carries replay command flit
- **Layer 2: Link Layer**
 - Supports 24 bit fabric addresses
 - Allows 10KB of L4 payload; 10,368 byte max packet size
 - Congestion Management
 - Adaptive / Dispersive Routing
 - Explicit Congestion Notification
 - QoS support
 - Traffic Class, Service Level, Service Channel and Virtual Lane
- **Layer 3: Data Link Layer**
 - Fabric addressing, switching, resource allocation and partitioning support



Courtesy: [Intel Corporation](#)

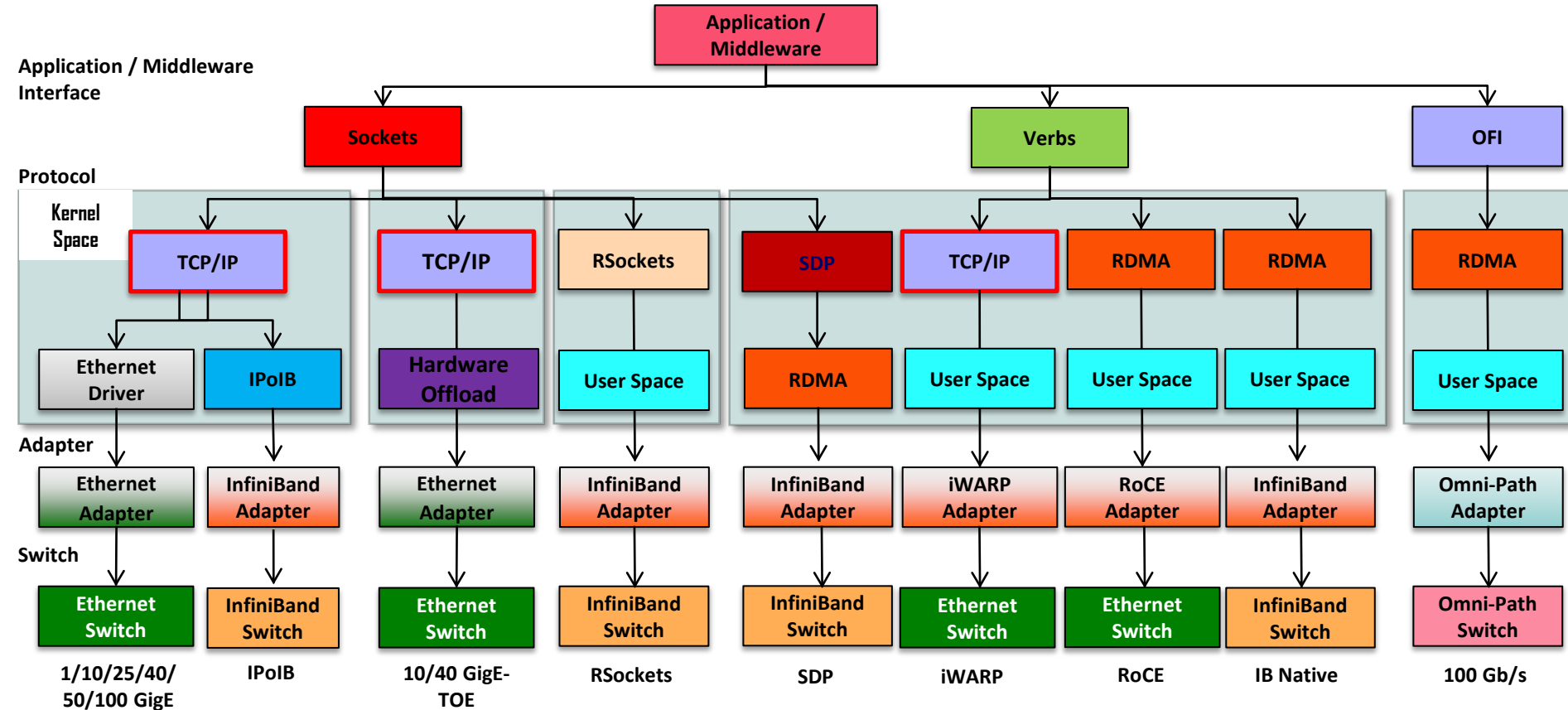
Layer 4: Transport Layer

- Multiple interfaces to access hardware capabilities
 - Performance Scaled Messaging (PSM)
 - Designed for HPC needs
 - Open Fabrics Interface (OFI)
 - AKA – Libfabrics
 - General Purpose APIs
 - Can be used by multiple vendors and L4 protocols
 - Open Fabrics Alliance Verbs
 - Designed for RDMA I/O



Courtesy: [Intel Corporation](https://www.intel.com)

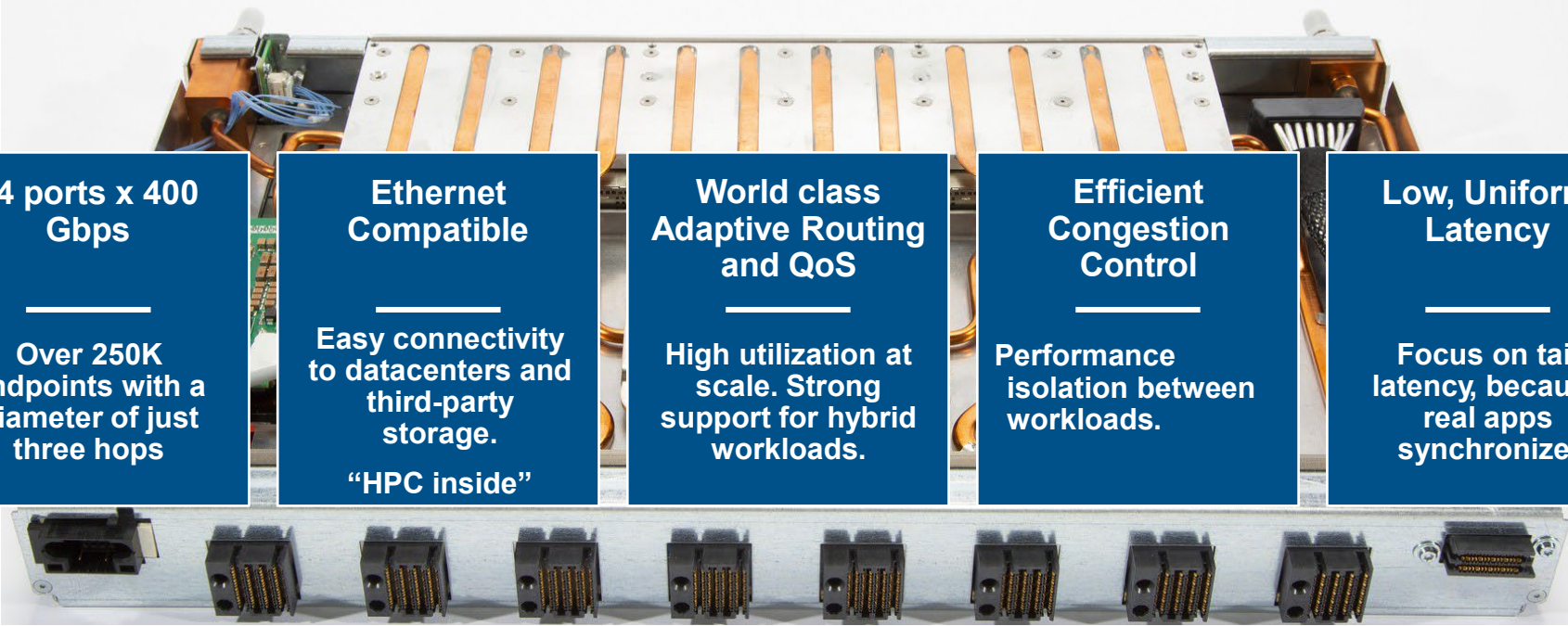
All Protocols Including Omni-Path



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Slingshot: HPE/Cray's 8th Generation Scalable Interconnect



64 ports x 400 Gbps

Over 250K endpoints with a diameter of just three hops

Ethernet Compatible

Easy connectivity to datacenters and third-party storage.
“HPC inside”

World class Adaptive Routing and QoS

High utilization at scale. Strong support for hybrid workloads.

Efficient Congestion Control

Performance isolation between workloads.

Low, Uniform Latency

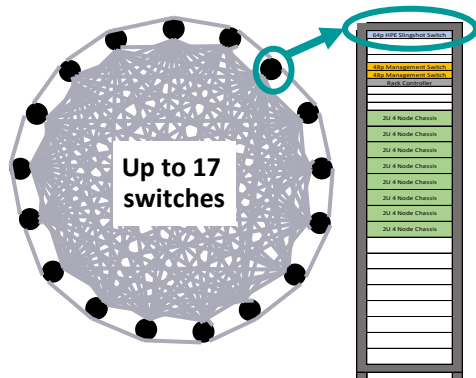
Focus on tail latency, because real apps synchronize.

Courtesy: HPE/Cray Inc. (ExaComm '19 Keynote Talk by Steve Scott)

HPE Slingshot topologies

Full Mesh

Up to 32 endpoints per switch

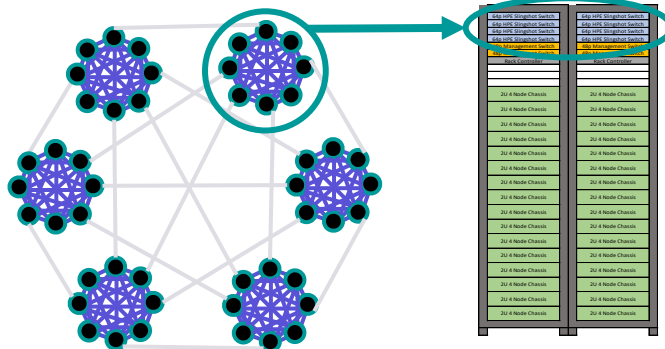


- Up to 17 fully connected switches
- Each 32 NICs up to 544 total
- Very cost-effective for clusters
- One hop minimal path

Dragonfly

Up to 16 endpoints per switch

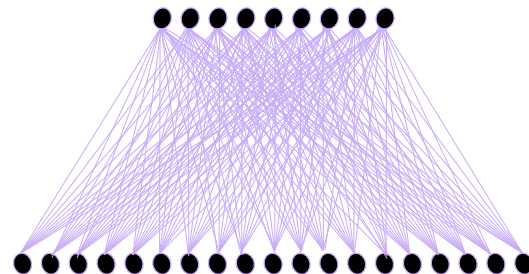
Illustrated: 8 switch group → 128 endpoints



- All-to-all group connects to every group
- Maximize copper for lower cost, less power
- Group size optimizes for cost, scale, deployment
- ≤ 3 switch-to-switch hops up to 250K endpoints
- Best performance for larger systems

Fat Tree

Mid-2026 FCS



- Easier to continually add capacity
- Cost advantage possible between 544 and 1024 endpoints and lower-performance use cases

Enabled by HPE Slingshot's unique networking technologies

Courtesy: <https://www.hpe.com/psnow/doc/a00150347enw>

A new option

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Ultra Ethernet Consortium

- Aimed at addressing the challenges posed by modern AI and HPC Jobs
- Aiming to build on top of advantage ethernet has in terms of adoption
- Goal -> “Tail latency” should be minimized
 - Multi-pathing and packet spraying
 - Flexible delivery order
 - Modern congestion control mechanisms
 - End-to-end telemetry
 - Larger scale, stability, and reliability
- Aims to address issues with current transport protocol services used by RoCE and IB ^[1]
 - Issues with DCQCN congestion control mechanism
 - Recovering from lost or out of order packet
 - Use a more scalable transport protocols with compared to RC which has N^2 connection overhead
 - Improved load balancing capabilities to handle larger messages/flows used by AI workloads
- Read the standard here: <https://ultraethernet.org/wp-content/uploads/sites/20/2026/08/UE-Specification-1.0.3.pdf>



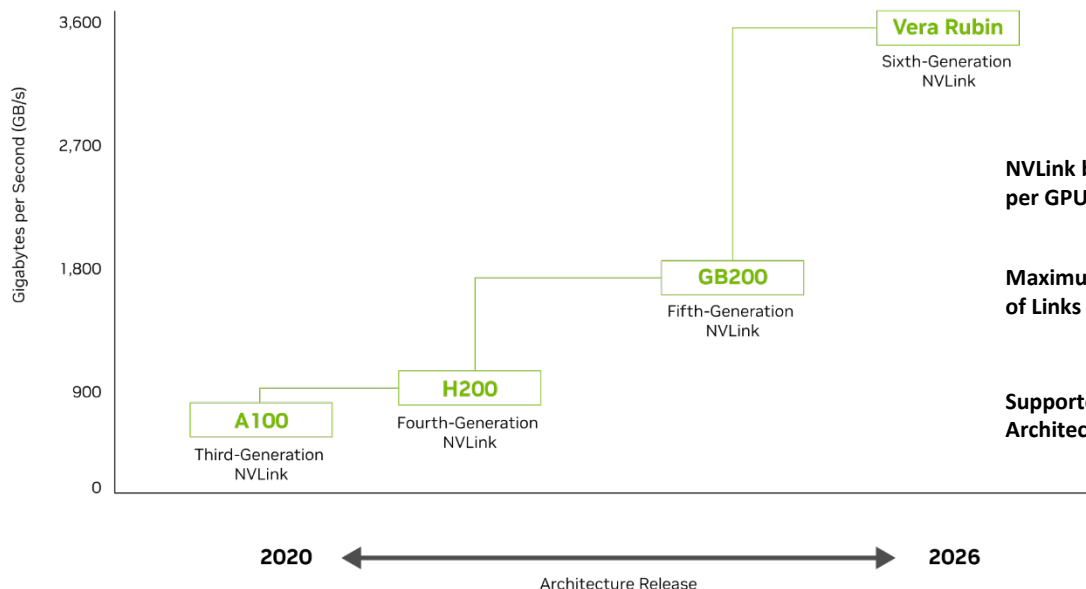
[1] Data Center Ethernet and Remote Direct Memory Access: Issues at Hyperscale, Hoefler et al., in Computer, July 2023

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NVLink (6th generation)

- High-performance interconnect for emerging dense GPU systems
 - Allows Load-Store operations between all GPUs



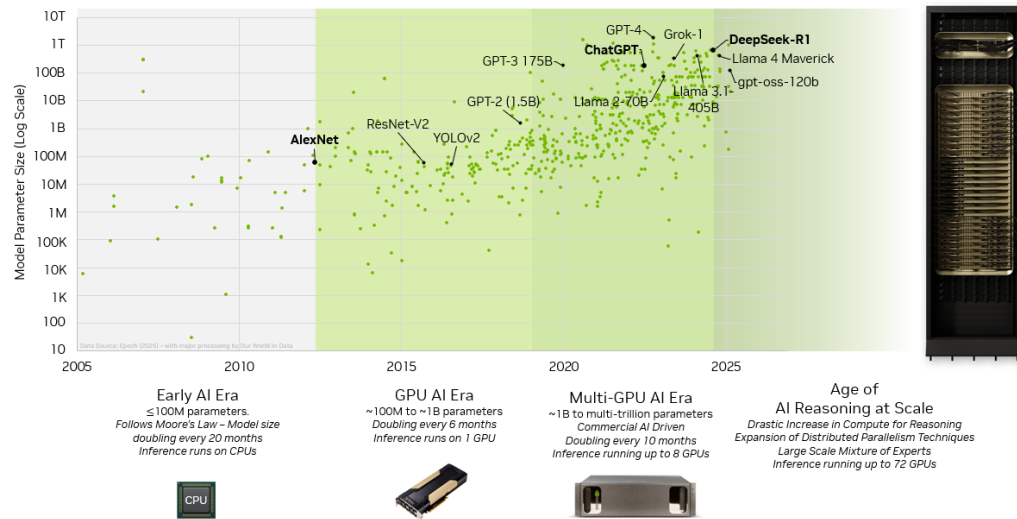
	Third Generation	Fourth Generation	Fifth Generation	Sixth Generation
NVLink bandwidth per GPU	600GB/s	900GB/s	1,800GB/s	3,600 GB/s
Maximum Number of Links per GPU	12	18	18	36
Supported NVIDIA Architectures	NVIDIA Ampere architecture	NVIDIA Hopper™ architecture	NVIDIA Blackwell architecture	NVIDIA Rubin Platform

NVLink Performance Trends

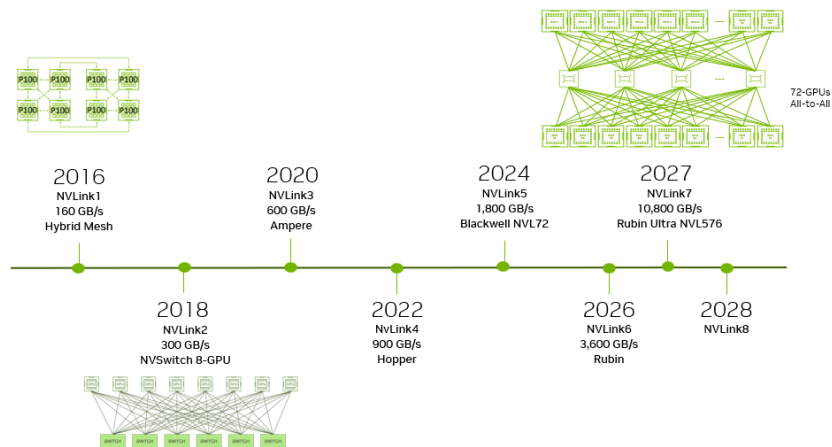
Courtesy: [NVIDIA](#)

Evolution of NVLink and NVSwitch

- Multiple GPUs on one node (Tesla V100s) → DGX All-to-all (8 GPUs/node) → Scale-up within NVL 4/72/576!
- Rack Scale Topology to offer advanced compute/communication capabilities
- Advanced All-to-all connectivity for 72 GPUs to enable multi-trillion-parameter training, inference, and AI reasoning



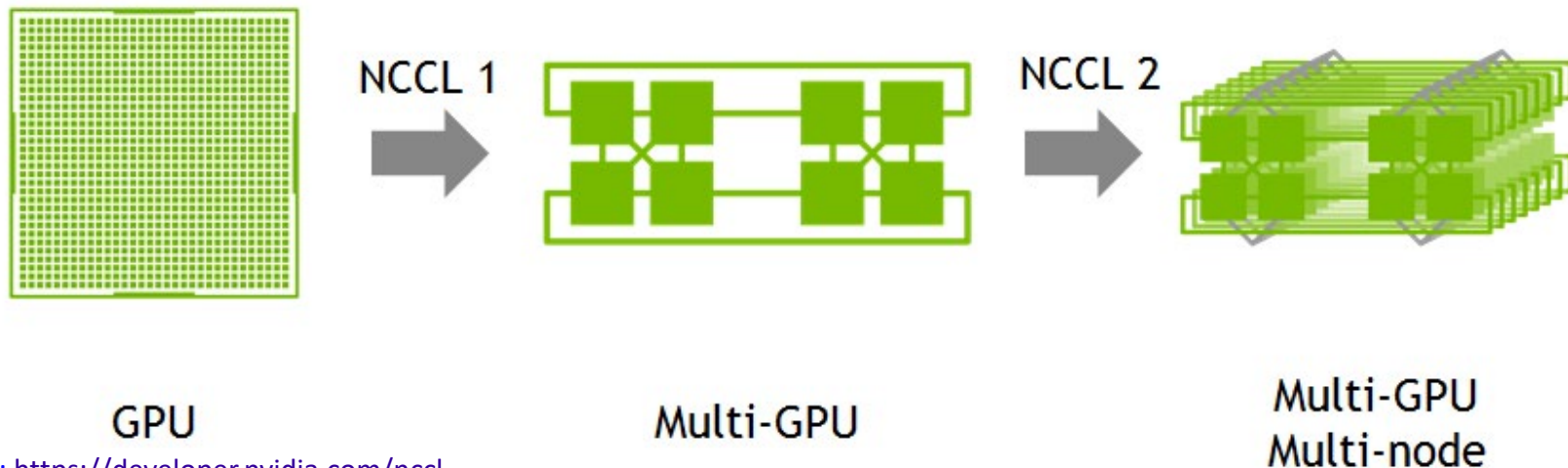
NVLink - Scale-Up Maturity with Rapid Innovation Pace



Courtesy: [Scaling AI Inference Performance and Flexibility with NVIDIA NVLink and NVLink Fusion](#)

NCCL Communication Library

- NVIDIA Collective Communication Library (NCCL)
- Main Motivation: Deep Learning workloads
- NCCL1– efficient dense-GPU communication within the node
- NCCL2– multiple DGX systems connected to each other with InfiniBand systems
 - NCCL 2.30.7: GPU-initiated communication (GIN), Elastic Buffers for resource management, etc.



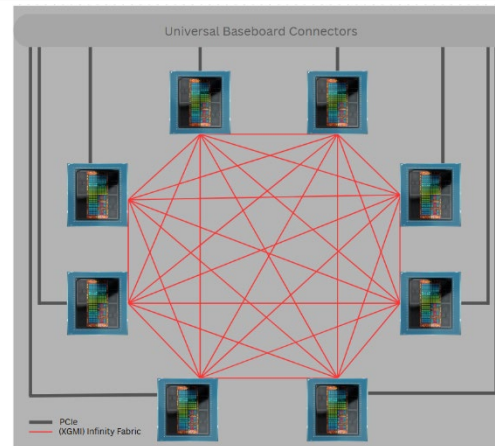
Courtesy: <https://developer.nvidia.com/nccl>

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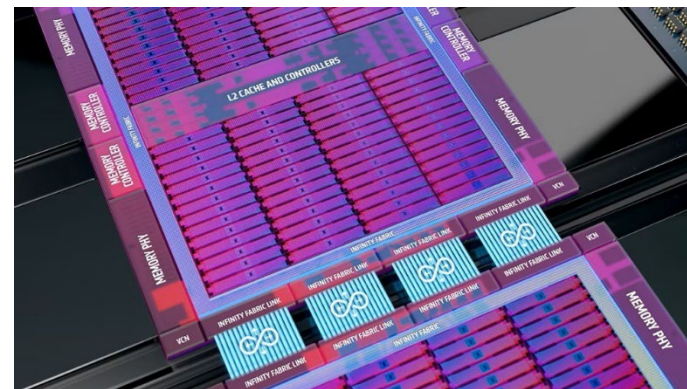
AMD Infinity Fabric/xGMI

- A cache coherent interconnect data fabric
- No switches – purely point-to-point framework
- Used within/across chiplets on CPUs as well as to provide a high-speed fabric between GPUs
- 64 GB/s per GPU-pt2pt link unidirectional BW
 - 128 GB/s bi-directional
- Theoretical 448 GB/s max unidirectional BW in the context of 8 MI300X GPUs
- Maintains cache coherence between CPUs and GPUs when using 3rd + generation AMD EPYC processors



Die to Die interconnect on CPUs

Courtesy: <https://rocm.blogs.amd.com/software-tools-optimization/mi300x-rccl-xgmi/README.html>

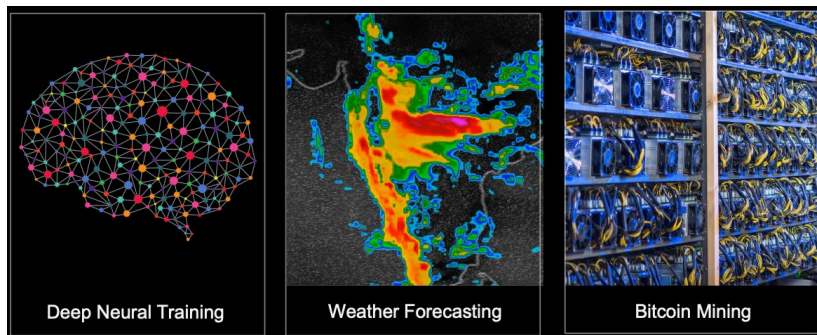


Infinity fabric

Courtesy: <https://www.amd.com/en/technologies/cdna2>

RCCL Communication Library

- ROCm Collective Communication Library (RCCL) – (pronounced “Rickel”)
- Uses the same C API as NCCL
- Intra-node communication support
 - PCIe and xGMI high-speed interconnects
 - InfiniBand, RoCE, and TCP/IP for inter-node communication.
- Useful for multi-GPU computing of workloads
 - Deep Neural training, Weather Forecasting, Bitcoin Mining



Multi-GPU computing use cases

Courtesy :

<https://www.amd.com/system/files/documents/multi-gpu-6.pdf>

<https://rocm.docs.amd.com/projects/rccl/en/develop/>

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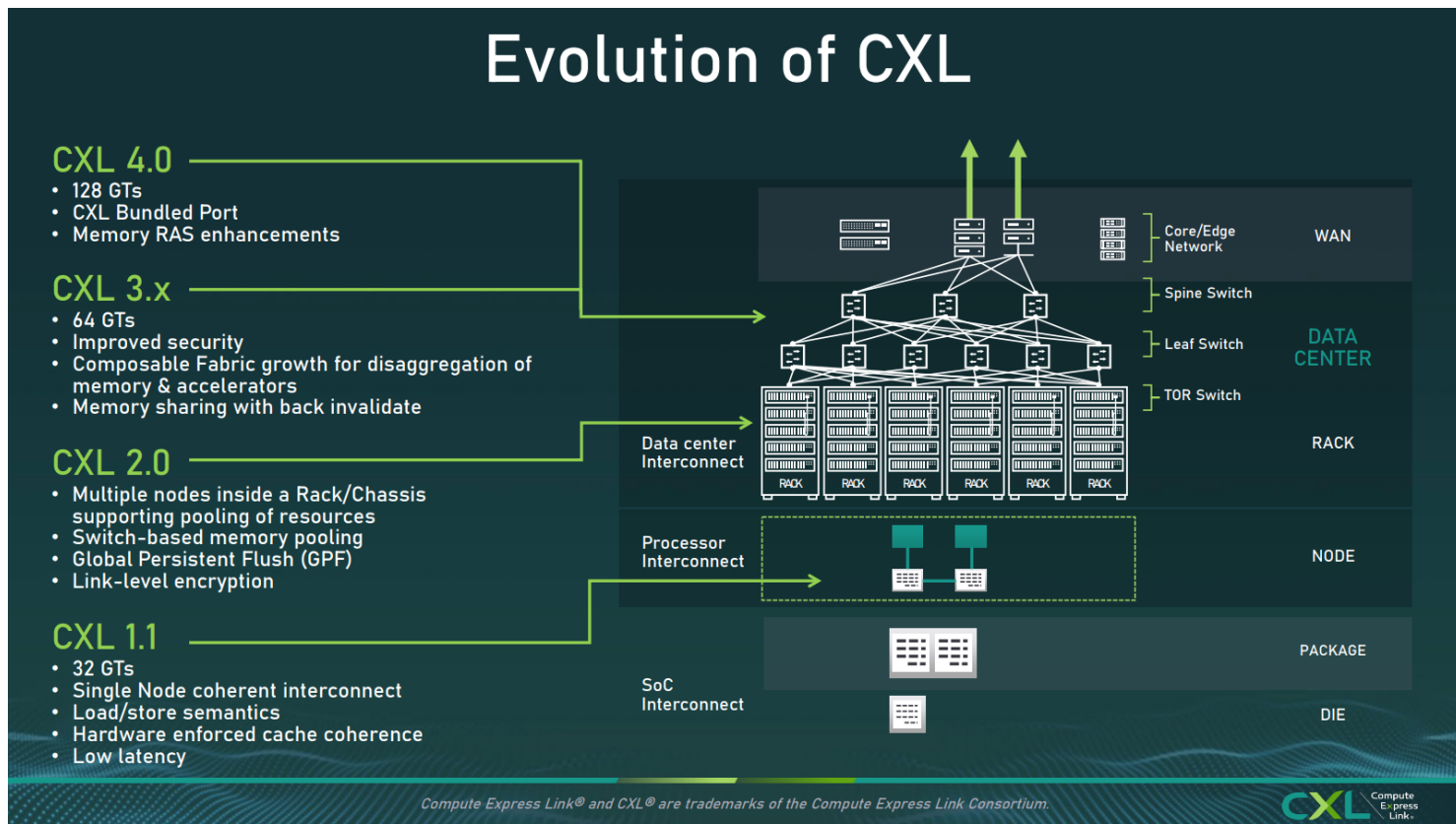
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Compute eXpress Link (CXL)

- Open industry standard
- Provides a cache coherent interconnect between
 - CPUs
 - Accelerators, like GPUs
 - Smart I/O devices, like DPUs, and
 - Various flavors of DDR4/DDR5 and persistent memories
- Allows the CPU to work on the same memory regions as the connected devices
- Improving performance and power efficiency while reducing software complexity and data movement

Courtesy: Toms Hardware & CXL Consortium

CXL Evolution



Courtesy: [CXL Consortium](https://cxlconsortium.org/)

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Ultra-Accelerator Link Consortium (UALink)



Courtesy:

<https://ualinkconsortium.org>

- Open Standard for AI Accelerator-to-Accelerator Communication
 - Abstract the notion of Accelerator to permit easy plug-and-play alongside interconnects such as CXL, PCIe, InfinityFabric/xGMI, etc.
- Focus on direct load, store, atomic ops between accelerators, connected via an “UltraLink Switch”
 - Low-latency/high-bandwidth fabric
 - 100s of accelerators supported within a pod
- Version 1.0: 200G support for multiple accelerator
- Version 2.0: Adds In-Network-Collective (INC) support, separate specification docs for modularity.
- Specifications available here: <https://ualinkconsortium.org/specification/>

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Broadcom Scale-Up Ethernet (UEC-Compliant Hardware with Tomahawk Ultra)

- Generic XPU-XPU interface to treat any accelerator or CPU as an endpoint for scale-up connectivity
- Use of an AXI target interface for control → supports 10K packets per second via unreliable transport.
- Includes an “SUE Lite” profile to reduce SUE IP by up to 50% (removes reliable transport layer and congestion control)

Table 2: SUE and SUE Lite Comparison

Attributes	SUE	SUE Lite
Transaction types	Write (Full/Partial, Posted/Non-Posted), Read, Message, Barrier	
Transaction size	256B	
DMA ready interface	Yes (Block write)	
Transaction packing	Yes	
LLR, CBFC, AFH Headers	Yes	
Core side interface	AXI4, Signal-based interface	Signal-based interface
End-to-end reliability	Yes (Reliable Transport Layer)	No (Hop-by-hop LLR only)
Congestion control	Yes (Fixed Window)	No
Partition Feature	Yes	No (Use address separation)

Courtesy: <https://docs.broadcom.com/doc/scale-up-ethernet-framework>

Figure 3: SUE Stack

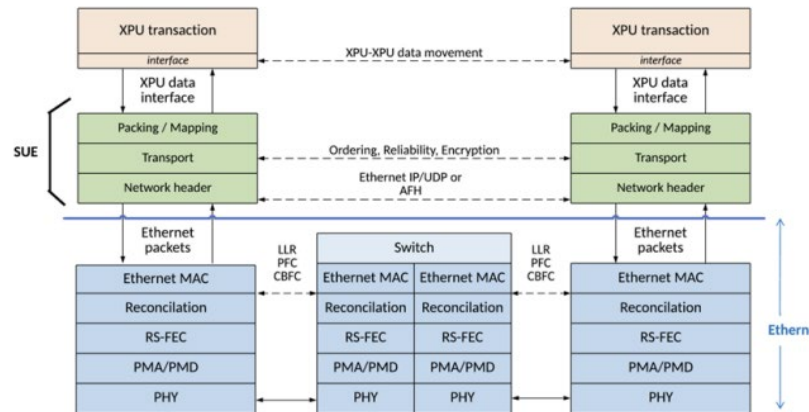
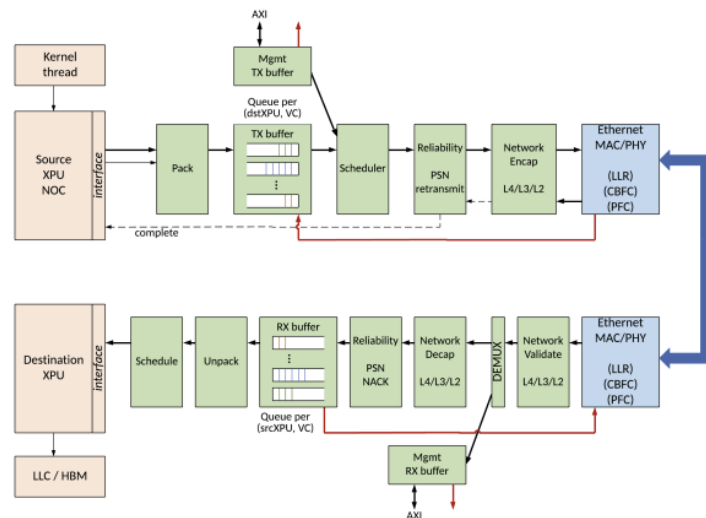
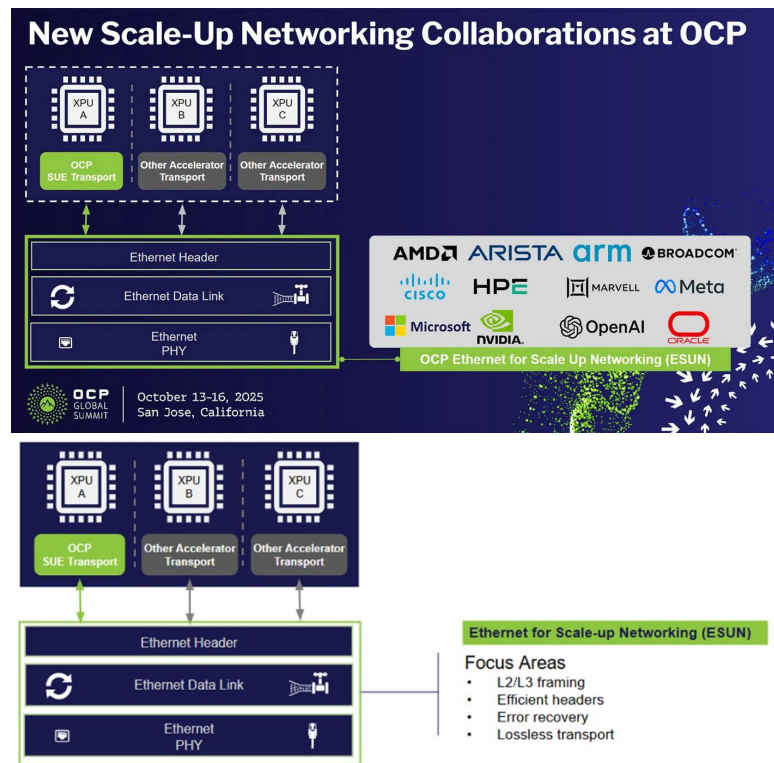


Figure 18: SUE Processing Flow



Open Compute Project: Ethernet for Scale-Up Networking (ESUN)

- October 2025 Keynote
- ESUN: Multi-Company Collaboration to enhance Ethernet for AU systems
- Focuses on L2/L3 (framing and switching) for lossless, error-resilient topologies.
- Works in conjunction with UEC



Courtesy:

<https://www.linkedin.com/feed/update/urn:li:activity:7356685219602579457/>;

<https://www.opencompute.org/blog/introducing-esun-advancing-ethernet-for-scale-up-ai-infrastructure-at-ocp>

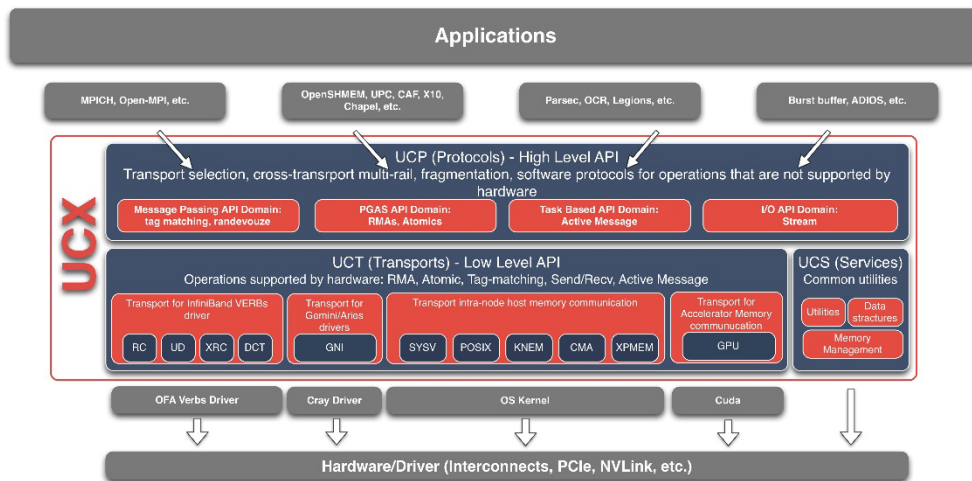
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Software Convergence with OpenFabrics

- Open-source organization (formerly OpenIB)
 - www.openfabrics.org
- Incorporates both IB, RoCE, and iWARP in a unified manner
 - Support for Linux and Windows
- Prior to OpenFabrics Interface (OFI), there was OpenFabrics Enterprise Distribution (OFED)
 - Latest stable release is OFED 4.17-1
 - New naming convention to get aligned with Linux Kernel Development
 - OFED 5.3 was under development until 2019, currently deprecated → OFI is the current active development with support for multiple network “providers”

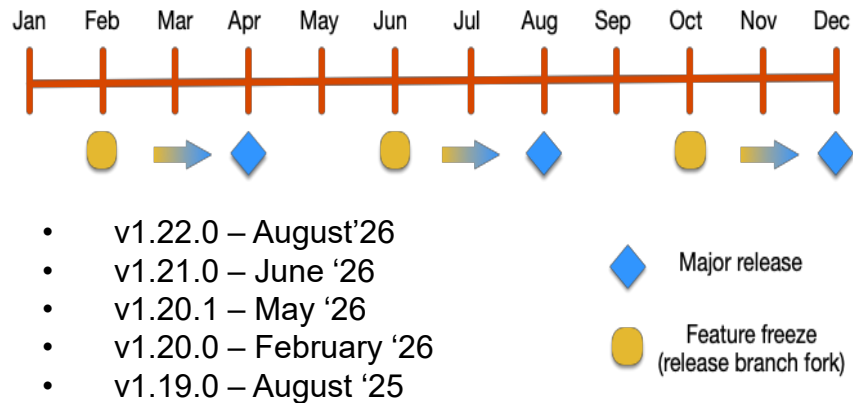
UCX Software Stack



- Collaboration between industry, laboratories, and academia to create production grade communication frameworks and open standards for data centric, ML/AI, and high-performance applications
 - Tuned Support for all CPU types (x86_64 Intel and AMD, ARM, IBM POWER 8/9 series)
 - Support for AMD and Nvidia GPUs
 - Runs on Servers, Raspberry PI like platforms, SmartNIC, Nvidia Jetson platforms, etc.

Courtesy: <https://www.openucx.org/>

UCX annual release schedule

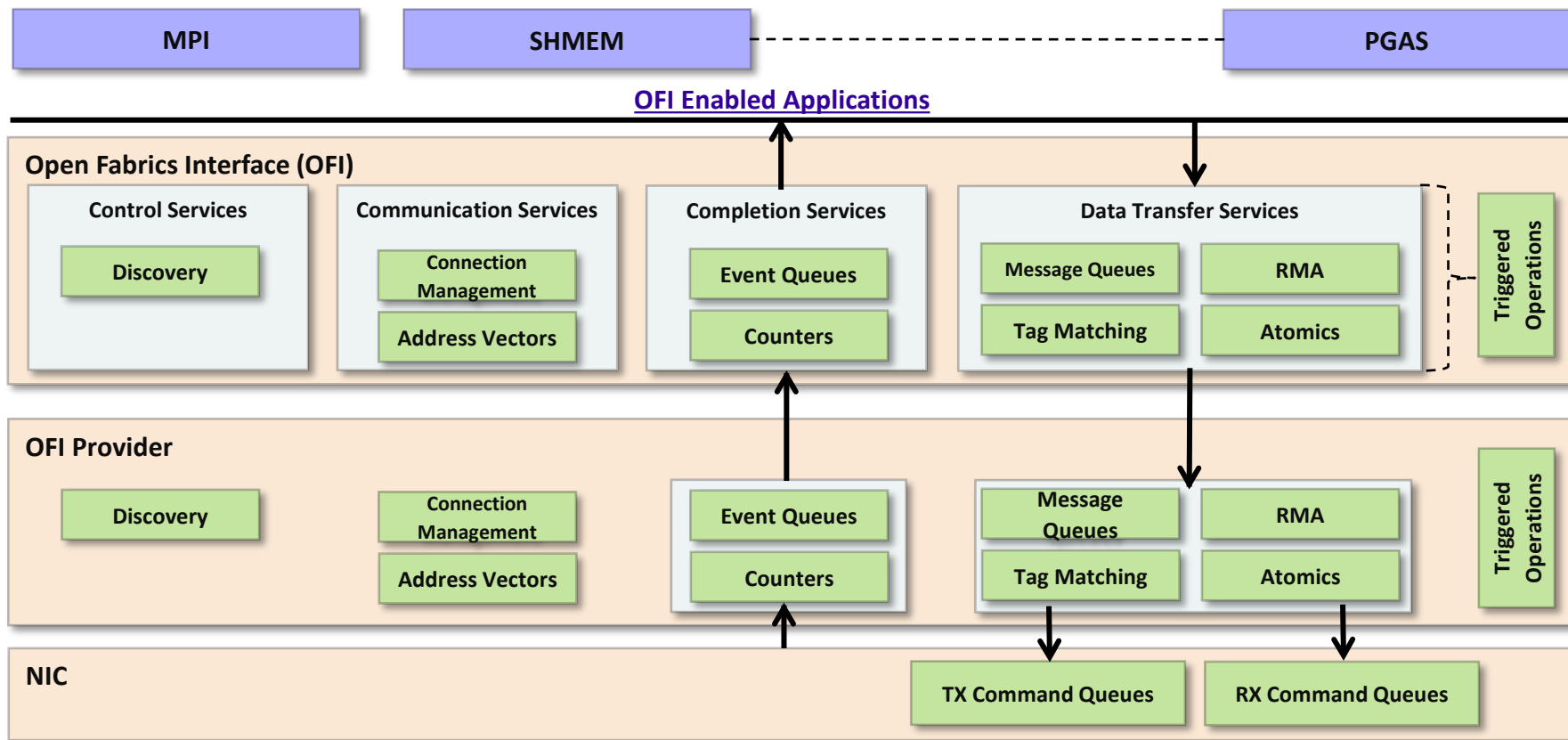


- v1.22.0 – August '26
- v1.21.0 – June '26
- v1.20.1 – May '26
- v1.20.0 – February '26
- v1.19.0 – August '25

Projects & Working Groups

- UCX – Unified Communication X
- UCC – Collective Library
- OpenSNAPI – Smart network Project
- SparkUCX – www.sparkucx.org
- UCD – Advanced Datatype Engine
- HPCA Benchmark – Benchmarking Effort

Libfabric Software Stack



Courtesy: <https://libfabric.org>

OpenFabrics Interface (OFI) Libfabric Release List Highlights

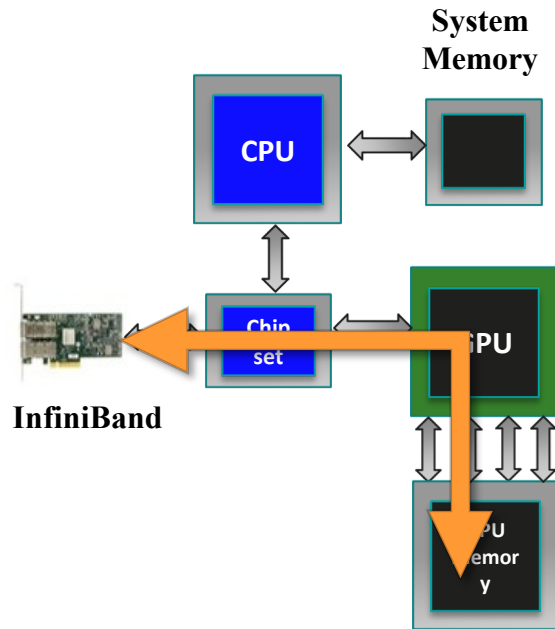
- <https://github.com/ofiwg/libfabric>
- OFI 1.0.0 (May 3, 2015): PSM, Sockets, usNIC, and IB “provider” support
- OFI 1.2.0 (January 7, 2016): Added GNI and PSM2 provider support)
- OFI 1.5.0 (August 9, 2017): Support for RxM (RC-like) and RxD (UD-like) transports
- OFI 1.8.0 (June 28, 2019): Support for Amazon EFA
- OFI 1.9.0 (November 22, 2019): Support for GPU-aware transports
- OFI 1.12.0 (March 8, 2021): Intel PSM3 support (Verbs supported, Unreliable Datagram interface)
- OFI 1.21.0 (March 22, 2024): Introduce initial upstreamed/open-sourced Slingshot provider support (“cxi”)
- OFI 2.0.0b (October 25, 2024): LINKx support (use of multiple providers simultaneously)
- Latest release: OFI Version 2.6.0 (June 19, 2026): Continued deprecation of FI_AV_MAP structure, multiple EFA-based enhancements

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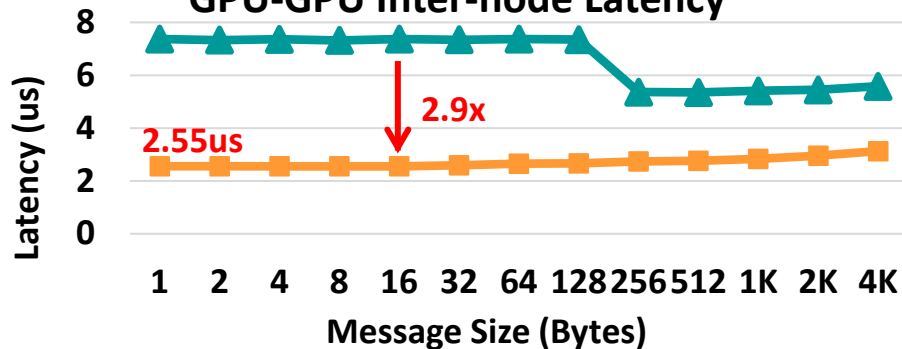
GPU-Direct RDMA

- Fastest possible communication between GPU and other PCI-E devices
- Network adapter can directly read/write data from/to GPU device memory
- Avoids copies through the host
- Allows for better asynchronous communication
 - Overlap of data movement and communication!
- Project done jointly between OSU, Mellanox, and NVIDIA during 2011-15. (ISC '11 paper on CUDA-Aware MPI)
- Very widely used in current days HPC and AI middleware currently for all GPU-based systems (NVIDIA, AMD, and Intel) with different interconnects (InfiniBand, Slingshot, Omni-Path, ROCE, etc.)

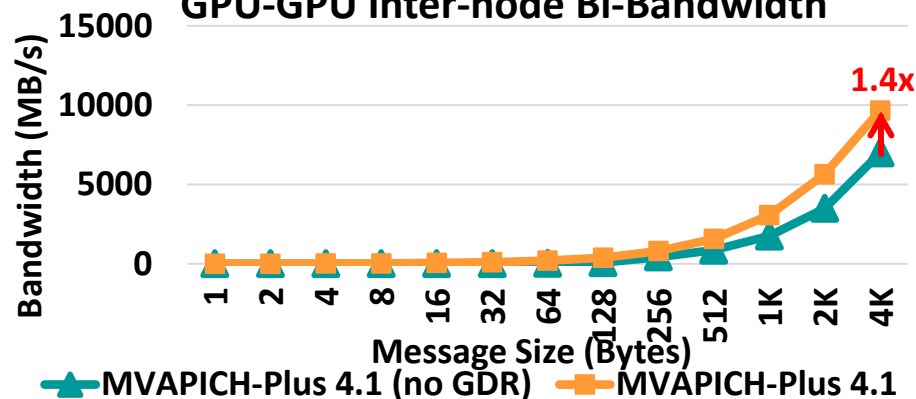


Optimized MVAPICH-Plus Design

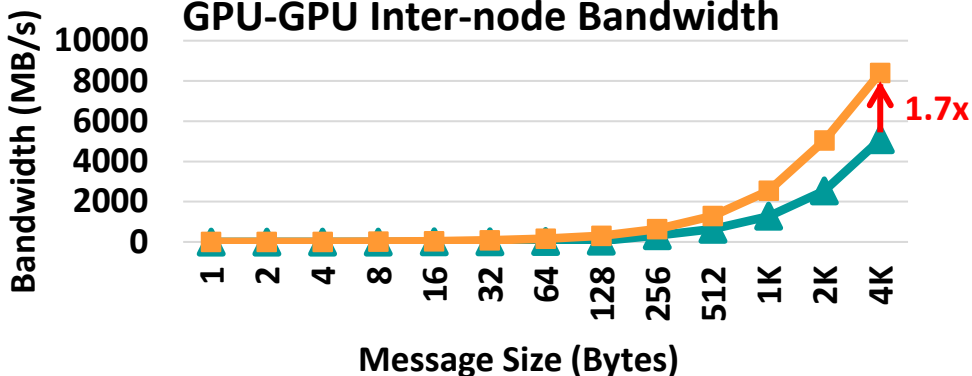
GPU-GPU Inter-node Latency



GPU-GPU Inter-node Bi-Bandwidth



GPU-GPU Inter-node Bandwidth



MVAPICH-Plus 4.1

GH200 Superchip (Grace @ 3.1 GHz) node - 72 cores

NVIDIA H100 GPU

Mellanox ConnectX-7 NDR400 HCA

CUDA 12.4

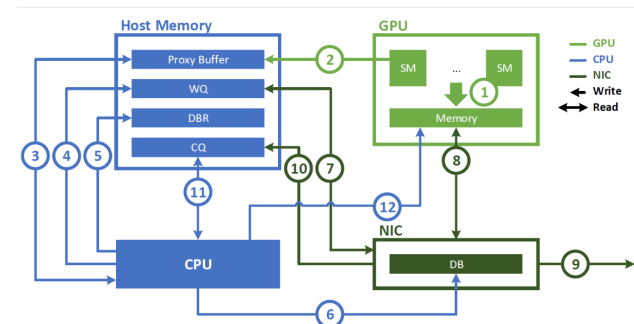
Mellanox OFED 25.01 with GPU-Direct-RDMA

MVAPICH-Plus 4.1 (no GDR) MVAPICH-Plus 4.1

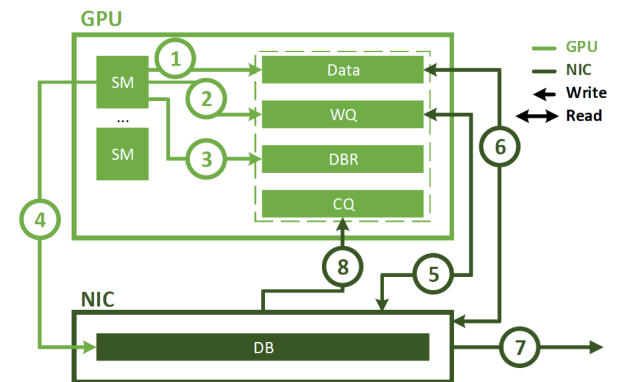
IB Integration with the GPU: IB GPUDirect Async (IBGDA)

- Normal GPU-based communication:
 - Proxy-thread initiated via the CPU to instruct GPU to perform RDMA transfers, CPU must notify the NIC to write to doorbell record
 - Incurs overhead with number of steps required (kernel-launching via proxy thread for data transfer), draws cycles away from compute, avoids peak NIC throughput, reduces goodput
- GPUDirect Async (GDA): Network-level primitives and setup to avoid crossing into the CPU on the critical path:
 - Direct control path from a GPU Streaming Multiprocessor (SM) to the NIC (launch a kernel from the CPU side, load GPU kernel, GPU kernel performs data transfer)
 - Async-kernel initiation (GPUDirect Async Kernel Initiation, GDAKI)
 - SM updates doorbell record and rings it without touching the CPU, polls for progression without CPU, ensures completion without CPU
 - Present in GPU-aware SHMEM libraries (NVSHMEM over IB, rocSHMEM over IB*), though requires small BIOS modifications to enable GPU-based shared memory/symmetric heap; Accessible through DOCA GPUNetIO library (UCX, NCCL, etc.

Proxy-initiated communication



InfiniBand GPUDirect Async



Courtesy: <https://developer.nvidia.com/blog/improving-network-performance-of-hpc-systems-using-nvidia-magnum-io-nvshmem-and-gpudirect-async/>

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Getting Set-up for the Hands-on Exercise 2 (GPU-Aware MPI library)

- You will run the experiments on the OSU RI2 cluster

- Please use the account name and password from

<http://go.osu.edu/ibtutorial>

- Open your favorite Terminal

\$ ssh ri2tut01@ri2.cse.ohio-state.edu

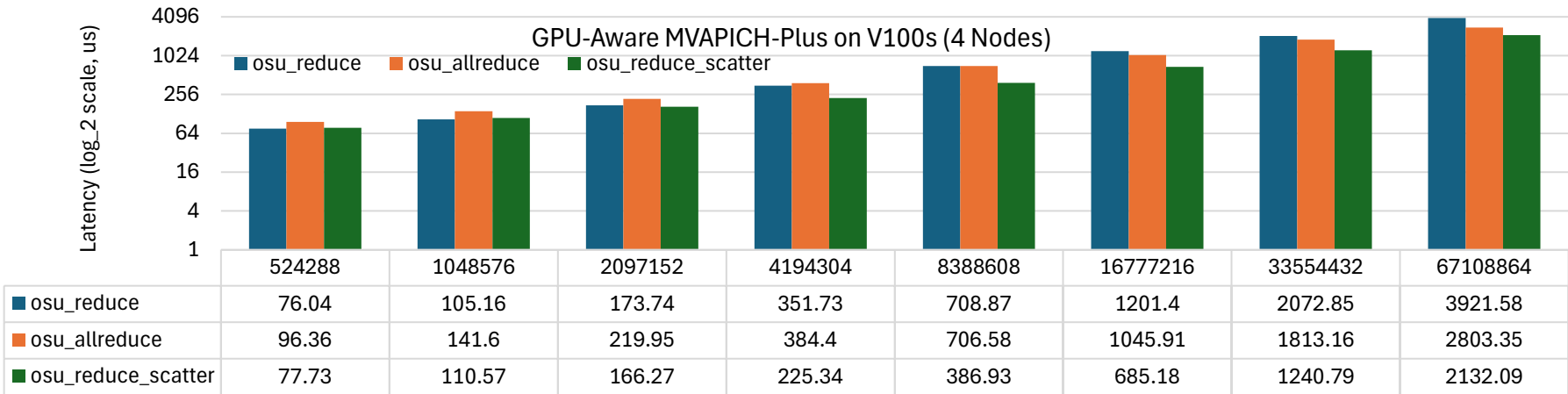
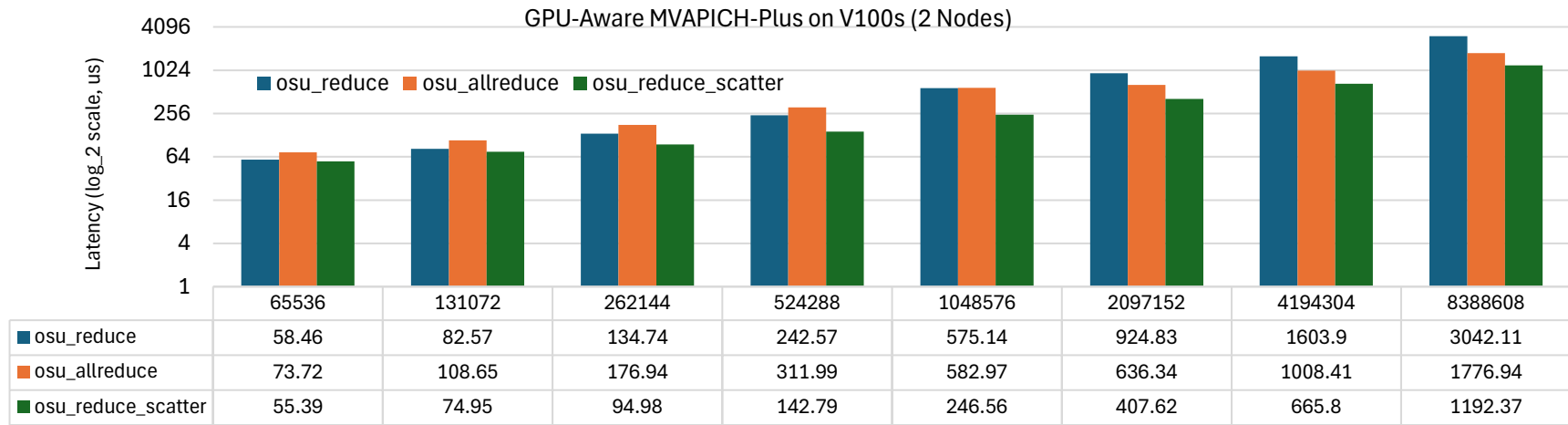
Enter Password:

- Make sure to `cd` to your home directory before cloning the repository
- sbatch -N <2 or 4> --reservation=ibtutorial
run_gpu.sh

- Clone from <https://github.com/OSU-Nowlab/HPN-Tutorial.git>
- Verify that the following files are present:

```
[michalowicz.2@ri2 HPN-Tutorial]$ ls -l
total 56
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 481 Jul 24 18:36 README.TXT
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 1801 Jul 27 13:02 run_dpu-rome.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 1379 Jul 27 13:01 run_dpu-thor.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 624 Jul 24 18:36 run_fabtest_bw.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 640 Jul 24 18:36 run_fabtest_lat.sh
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 1356 Jul 27 13:02 run_gpu.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 673 Jul 24 18:36 run_inline_ib.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 668 Jul 24 18:36 run_inline_mpi.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 683 Jul 24 18:36 run_mtu.sh
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 433 Jul 24 18:36 run_omb_ofi.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 444 Jul 24 18:36 run_omb.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 498 Jul 24 18:36 run_perftest_bw.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 501 Jul 24 18:36 run_perftest_lat.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 866 Jul 24 18:36 run_rc_ud.sh
```

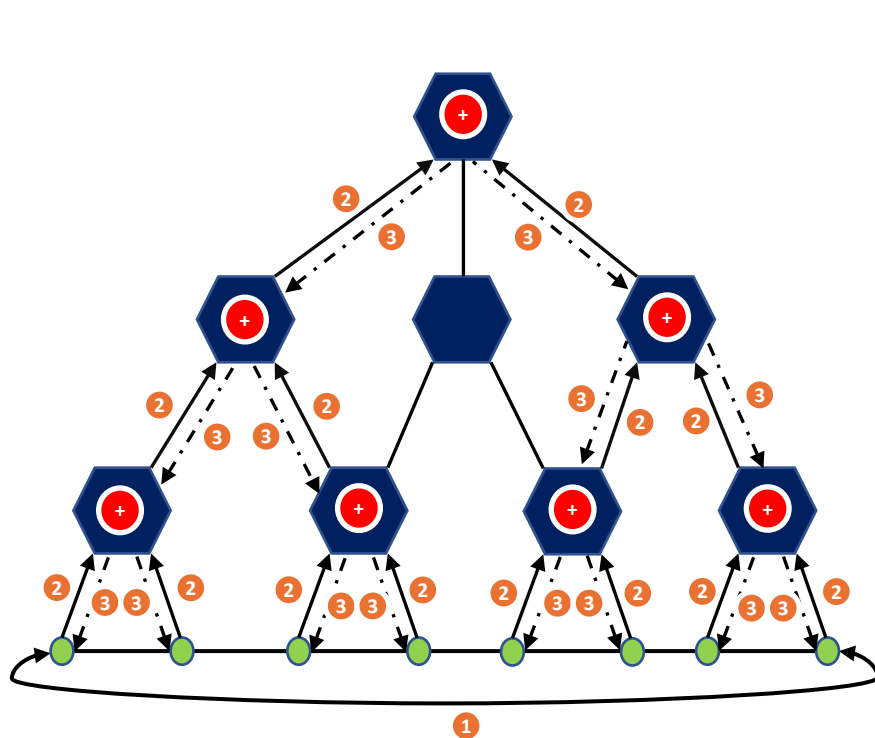
GPU-Aware collectives with MVAPICH-Plus 4.1 (In-house Cluster @ OSU)



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In-Network Computing - Collective Support (SHARP)



-  CPU/GPU
-  Reduce data in switch
-  Switch
-  Broadcast reduced data from switch
-  Send data to switch

- 1 Initialize in-network offload once per job
- 2 Send data to switch
- 3 Receive data from switch

No involvement of CPU/GPU once data reaches switches

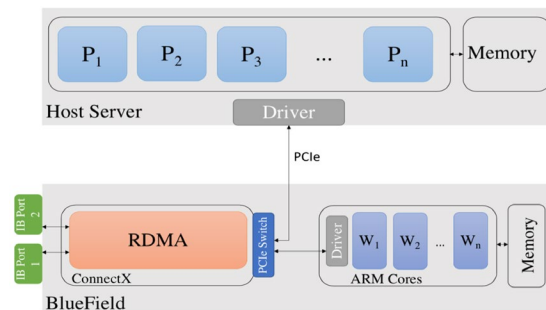
SmartNICs

- Broadly two types of SmartNICs
 - CPU based : NIC + Programmable CPU cores + ASICs
 - NVIDIA Bluefield Data Processing Units (DPUs)
 - Marvell Octeon
 - AMD Pensando/Pollara DSC
 - Intel EP2200 and related products
 - Field Programmable Gate Arrays (FPGAs) based : NIC + FPGAs
 - AMD Alveo
 - Intels FPGA SmartNICs



Evolution of NVIDIA BlueField DPU Subsystem Hardware

Metric	BlueField-2	BlueField-3	BlueField-4
Processor	ARM Cortex A-72	ARM Cortex A-78	ARM Neoverse-V2
Core Count	8	16	64
L1 Cache (per-core)	32KB D-cache, 48KB I-Cache	64KB I/D-Cache	64KB 4-way I/D-cache
L2 Cache (per-core)	1MB	512KB	1MB/2MB
L3 Cache (shared)	6 MB	16 MB	N/A
On-Chip RAM	16GB	32GB	128GB
Memory Controller Count	1	2	4
DRAM clock speed	1600 MT/s (DDR4)	5600 MT/s (DDR5)	6400 MT/s (LPDDR5)
Max Interconnect Speed	200 Gb/s	400 Gb/s	800 Gb/s



- Plus engines for encryption, compression/decompression, RISC-V datapath accelerator, etc.
- Courtesy: <https://resources.nvidia.com/en-us-accelerated-networking-resource-library/bluefield-4-dpu-datasheet>, <https://resources.nvidia.com/en-us-accelerated-networking-resource-library/datasheet-nvidia-bluefield> (BlueField-3), <https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/documents/datasheet-nvidia-bluefield-2-dpu.pdf>

MVAPICH2-DPU Library Release

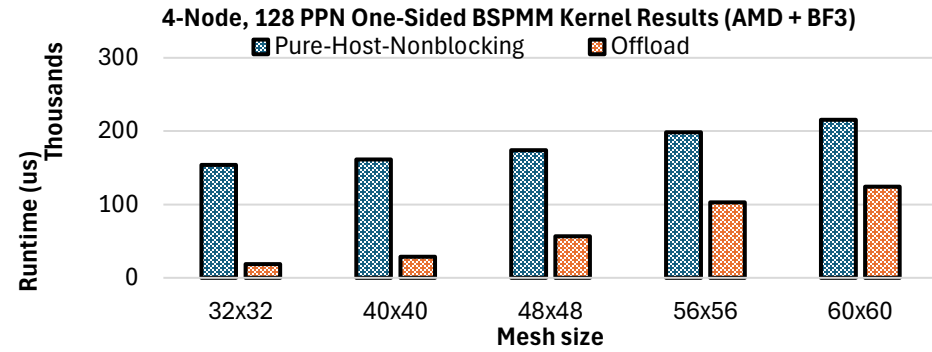
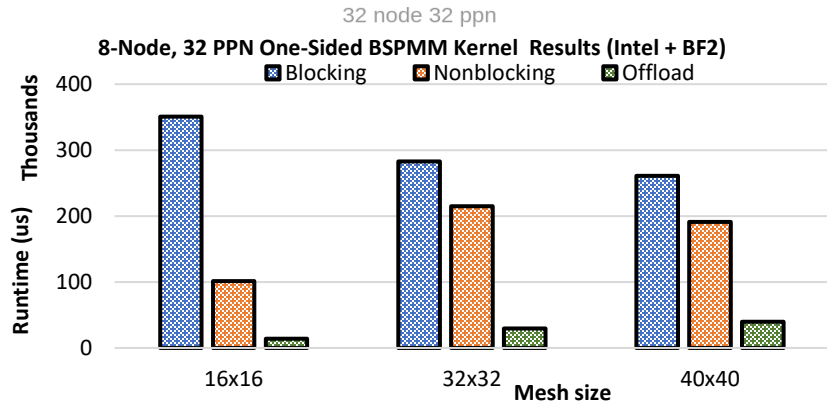
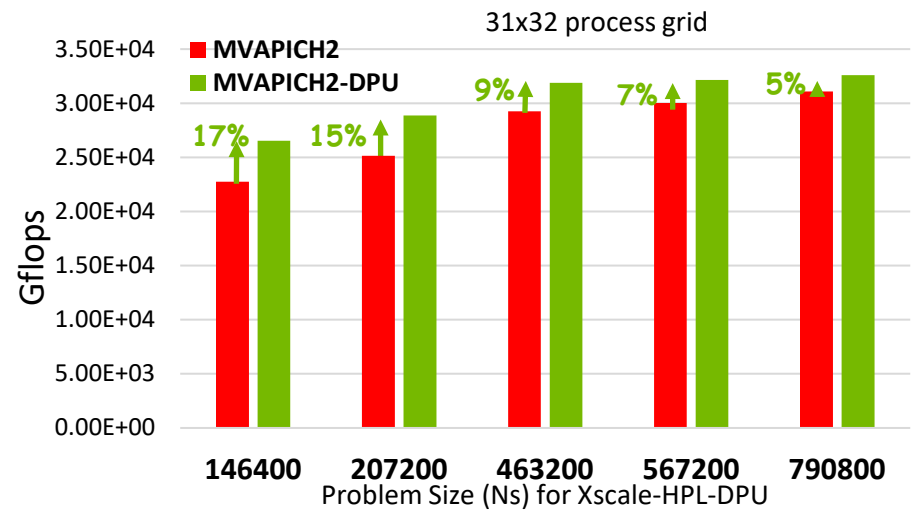
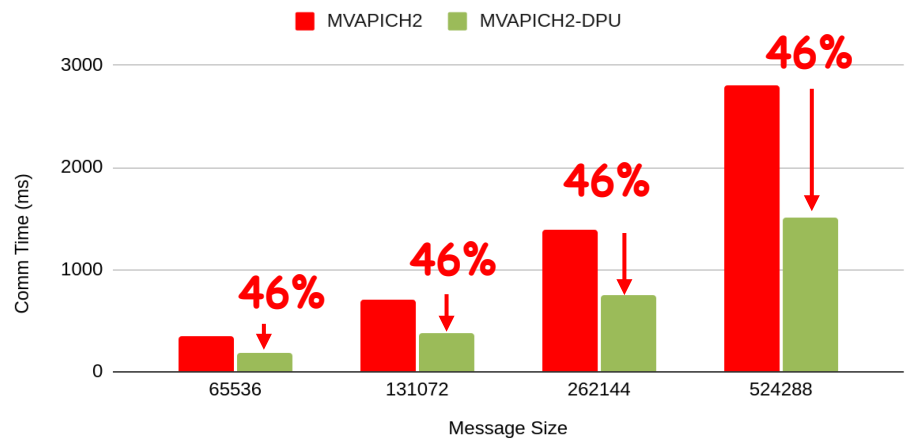


- Supports all features available with the MVAPICH2 release (<http://mvapich.cse.ohio-state.edu>)
- Novel framework to offload non-blocking collectives to DPU
- Offloaded functionality:
 - MPI_Ialltoall/v (nonblocking alltoall/v)
 - MPI_Ibcast (nonblocking bcast)
 - MPI_Iallgather (nonblocking allgather)
 - MPI_I/send, MPI_I/recv (non/blocking point to point)

Available from X-ScaleSolutions, please send a note to contactus@x-scalesolutions.com to get a trial license.

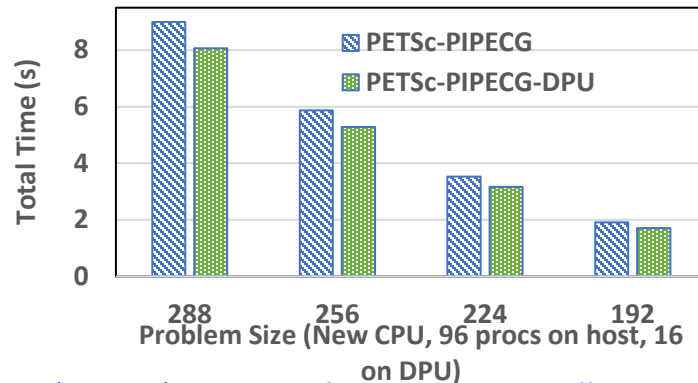
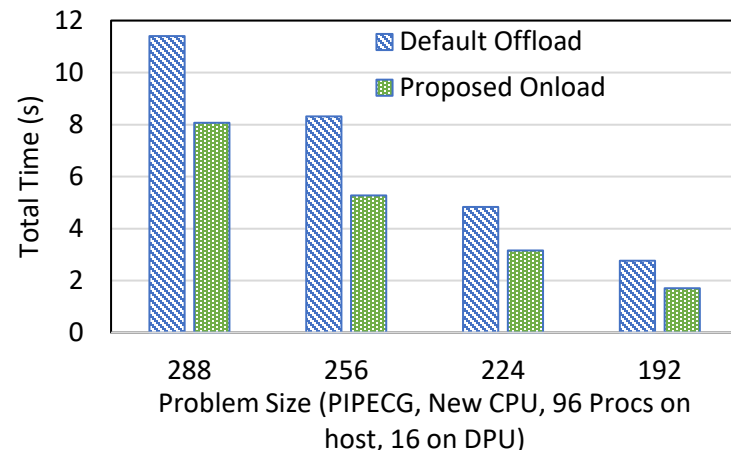
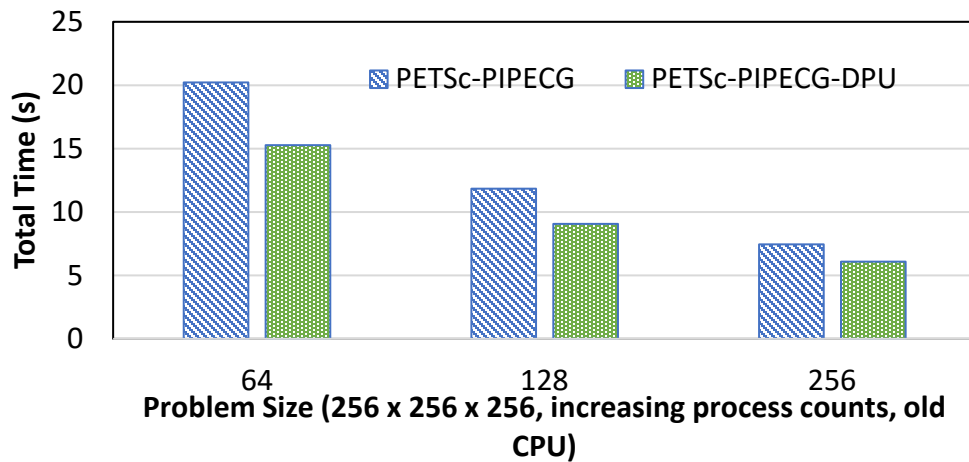
Subset of Expected Results from DPU Communication Offload (NVIDIA BlueField)

Total Execution Time BF-2 (osu_ialtoall)



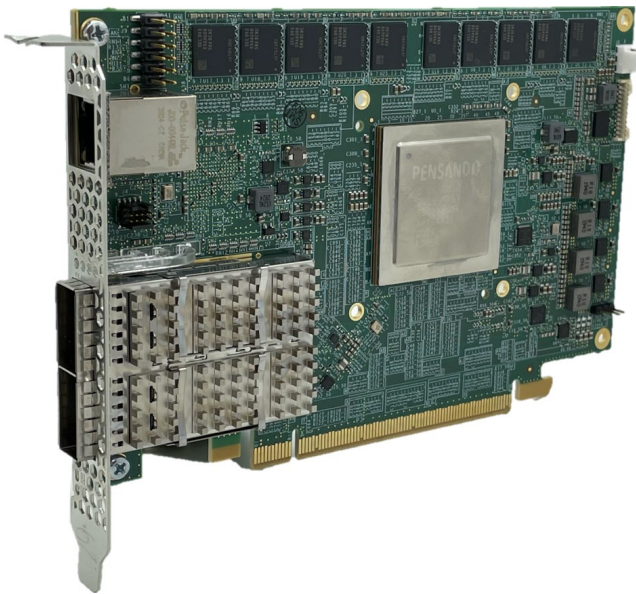
Smart Compute Offload (Hybrid CPU + BlueField-3 SmartNICs)

- Targeted towards libraries like PETSc and HYPRE
 - Creating a set of APIs for Vector-Multiply Add (VMA), Distributed Dot (DDOT), and Matrix-Vector (MATVEC) operations
 - Onloading Scheme for reducing cost of data movement
- Older CPUs (Intel Broadwell) + BF3: Up to 24% performance improvement
- Newer CPUs (Intel SPR) + BF3: Up to 10-15% Improvement



K. Suresh, B. Michalowicz, N. Contini, B. Ramesh, M. Abduljabbar, A. Shafi, H. Subramoni, and DK Panda, Using Bluefield-3 SmartNICs to Offload Vector Operations in the Kryolov Subspace Method, HiPC '24, Dec 2024.

AMD SmartNICs

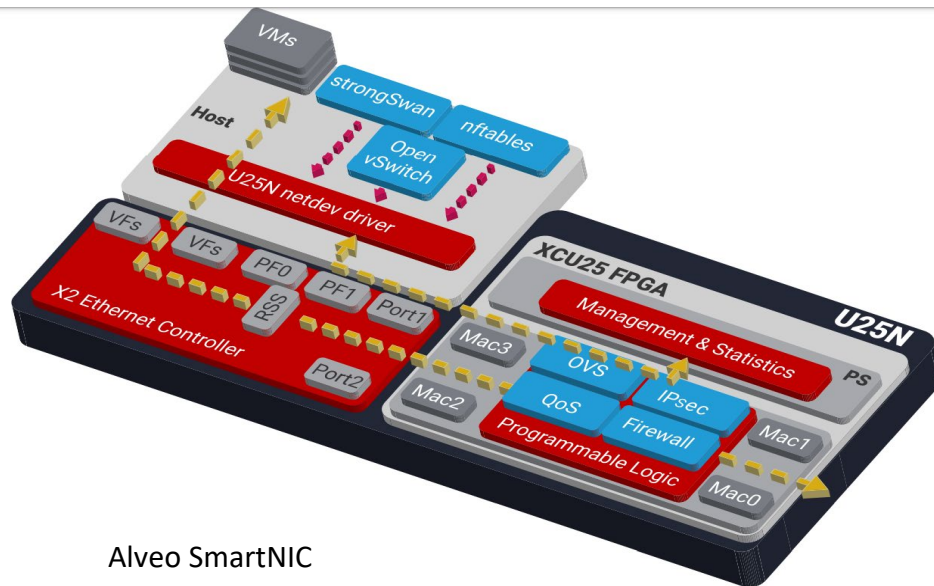


AMD Pensando DSC

- Powered by Pensando DPU – paired with frontend “Pollara” chip
- P4- programmable custom match processing units (MPUs)
- combined with a 16x A72 ARM® core complex
- dedicated data encryption and storage offload engines

Two types of SmartNICs

- CPU based AMD Pensando DSC
- FPGA based Alveo

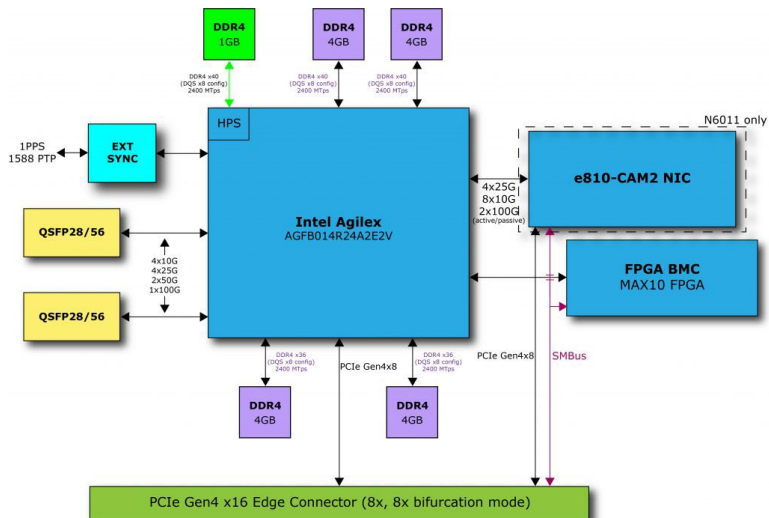


Alveo SmartNIC

- Eg: U25N SmartNIC
- XtremeScale™ X2 Ethernet Controller
- AMD UltraScale+™ FPGA
- Multi-core Arm processor
- FPGA has programmable dataplane pipelines like QoS, IPsec, Match Engine

Intel IPU and SmartNIC Examples

Intel FPGA SmartNIC NL6010/11



- 2x100 Gbps Ethernet
- Onboard Ethernet Controller
- Intel Agilex® FPGA
- Released 2024

Intel E2200 IPU/DPU



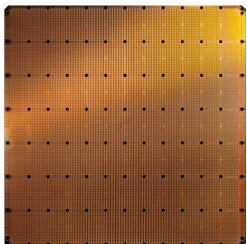
- P4 packet processing pipeline, 400G Bandwidth
- Inline Crypto Engine
- RDMA and storage capability including NVMe offload
- ARM Neoverse based compute complex
- Courtesy:
<https://www.servethehome.com/intel-ipu-e2200-400g-dpu-at-hot-chips-2025/>

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Cerebras Architecture

- CS1: Released in 2019 with 1.2 trillion transistors
- CS3 – released in 2024 with 4+ trillion transistors!
 - Released in March 2024
 - Over 4 trillion (!!) transistors
- Configured with up to 1.2 Petabytes of external memory
- 250 Exaflops of AI performance with 2048 CS-3 systems linked together
 - Train Llama2-70B in one day
- In use for high-intensity HPC workloads (molecular dynamics) as well!



Cerebras WSE-2

2.6 trillion transistors

46,225 nm²

Wafer Scale Engine-3 vs. H100

	Cerebras WSE-3	Nvidia H100	Cerebras Advantage
Chip size	46,225 mm ²	814 mm ²	57x
Cores	900,000	16,896 FP32 + 528 Tensor	52x
On-chip memory	44 Gigabytes	0.05 Gigabytes	880x
Memory bandwidth	21 Petabytes/sec	0.003 Petabytes/sec	7,000X
Fabric bandwidth	214 Petabits/sec	0.0576 Petabits/sec	3,715X

Spec	Cerebras CS-3	B200	DGX B200	GB200 NVL72	CS-3 / B200	CS-3 / DGX B200	CS-3 / NVL72
FP16 PFLOPs	125	4.4	36	360	28.4	3.5	0.3
Memory (GB)	1,200,000	192	1,536	13,500	6,250.0	781.3	88.9
NVLink Fabric Bandwidth (TB/s)	26,750	1.8	14.4	130	14,861	1,858	206
Power (Watts)	23,000	1,000	14,300	120,000	23.0	1.6	0.2
PFLOPs / W	0.005	0.004	0.003	0.003	1.2	2.2	1.8

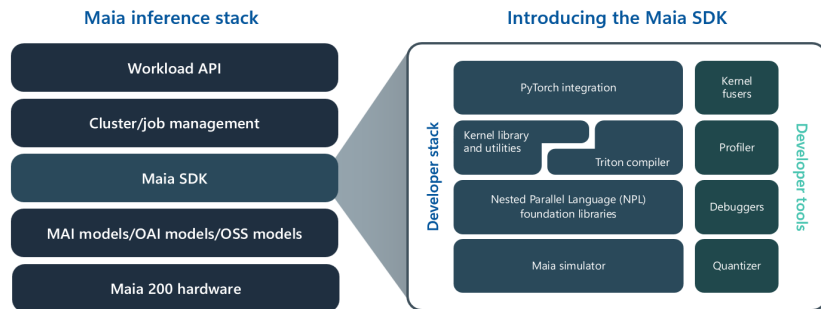
Microsoft Maia 200: “The AI Accelerator Built for Inference”

Industry-leading capability

- Microsoft HW for AI inference on in-house Azure deployments
- Features:
 - 272MB on-die SRAM, partitioned to Cluster and Tile-level channels; low-latency, high-bandwidth data management
 - Custom on-chip interconnect for high-throughput transfers (bcast, gather, reduce, scatter via Microsoft Collective Communication Library (MCCL)) and enhanced tile-to-tile SRAM access
 - Integrated NIC on Chip (NOC), QOS for critical traffic (think Service Lanes), Fault-tolerant Management Plane
 - Two-Tier ScaleUp Interconnect with proprietary “AI Transport Layer” – scales to 6144 Maia accelerators
- Maia SDK for software development

Peak specifications	Azure Maia 200	AWS Trainium3	Google TPU v7
Process Node	3nm	3nm	3nm
FP4 TFLOPS	10,145	2,517	n/a
FP8 TFLOPS	5,072	2,517	4,614
BF16 TFLOPS	1,268	671	2,307
HBM technology	HBM3E	HBM3E	HBM3E
HBM BW (TB/s)	7	4.9	7.4
HBM capacity (GB)	216	144	192
Scale-up BW: bidirectional (TB/s)	2.8 TB/s	2.2-2.56 TB/s	1.2 TB/s

Maia software stack



Courtesy: <https://techcommunity.microsoft.com/blog/azureinfrastructureblog/deep-dive-into-the-maia-200-architecture/4489312> ; <https://blogs.microsoft.com/blog/2026/01/26/maia-200-the-ai-accelerator-built-for-inference/>

Meta MTIA 3rd/4th/5th Generation Inference Chips

- Versions 1 and 2: 2023 and 2025 (See ISCA papers from those years)
- Made initially for Ranking and Recommendation (R&R) model training prior to GenAI
- MTIA 300/400/450/500: see increased scale-up network bandwidth
 - MTIA 400: 72-accelerator scale-up domain; on-path to deployment
 - MTIA 450: doubled HBM performance; to be deployed in early 2027
 - MTIA 500: 50% more HBM performance and mixed precision flops. To be deployed closer to 2028

Courtesy: <https://ai.meta.com/blog/meta-mtia-scale-ai-chips-for-billions/>

Metric	MTIA 300	MTIA 400	MTIA 450	MTIA 500
Workload Focus	R&R Training	General	GenAI Inference	GenAI Inference
Module TDP	800 W	1200 W	1400 W	1700 W
HBM Bandwidth	6.1 TB/s	9.2 TB/s	18.4 TB/s	27.6 TB/s
HBM Capacity	216 GB	288 GB	288 GB	384-512 GB
MX4 Performance	—	12 PFLOPs	21 PFLOPs	30 PFLOPs
FP8/MX8 Performance	1.2 PFLOPs	6 PFLOPs	7 PFLOPs	10 PFLOPs
BF16 Performance	0.6 PFLOPs	3 PFLOPs	3.5 PFLOPs	5 PFLOPs
Scale-up Domain Size	16	72	72	72
Scale-up Network (unidirectional bandwidth*)	1 TB/s	1.2 TB/s	1.2 TB/s	1.2 TB/s
Scale-out Network (unidirectional bandwidth*)	200 GB/s**	100 GB/s	100 GB/s	100 GB/s

*Some vendors report bidirectional bandwidth. Multiply the value in the table by two to obtain the corresponding bidirectional bandwidth.

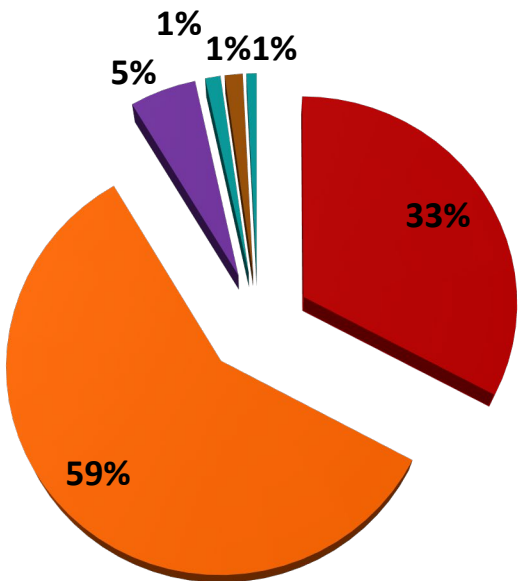
**MTIA 300 is configured with a scale-out network with higher bandwidth (200 GB/s) due to its relatively small scale-up domain size and the target R&R workloads.

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- **Network trends in the Top500**
- Sample and Performance Numbers
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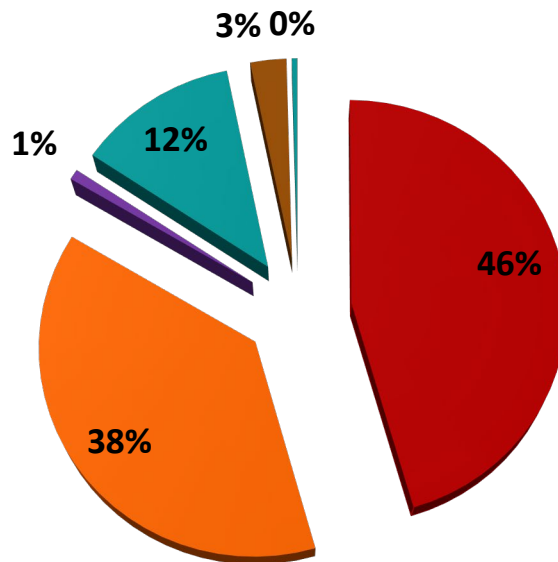
InfiniBand in the Top500 (June 2026)

Count



- Gigabit Ethernet
- Infiniband
- Omnipath
- Custom Interconnect
- Proprietary Network
- Ethernet

Performance



- Gigabit Ethernet
- Infiniband
- Omnipath
- Custom Interconnect
- Proprietary Network
- Ethernet

Large-scale InfiniBand Installations

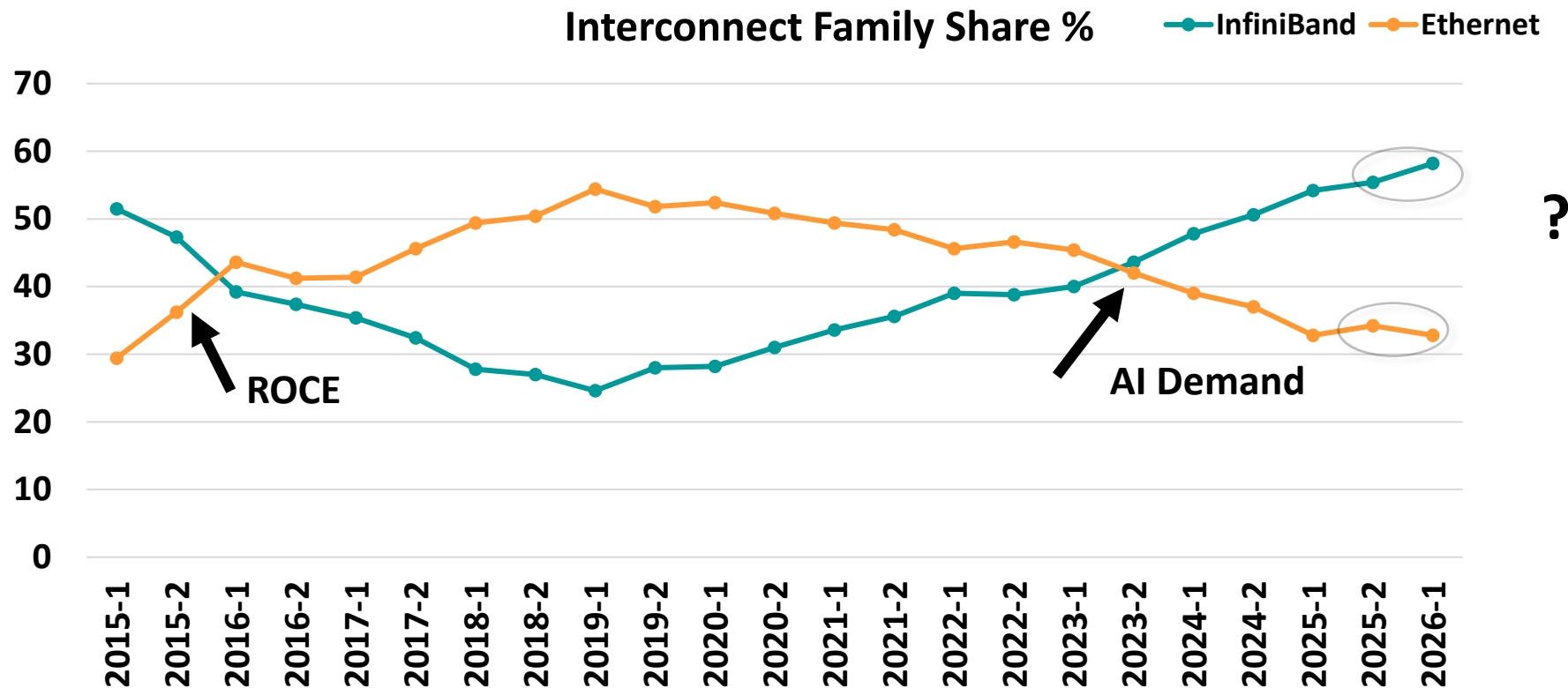
- 292 IB Clusters (58.2%) in the June 2026 Top500 list
 - (<http://www.top500.org>)
- Some of the installations in the Top 50 (22 systems):

4,801,344-core (JUPITER Booster) @ Eviden/EuroHPC/FZJ in Germany (5th)	297,840-core (CHIE-2),@Softbank in Japan (30 th)
1,123,200-core (Eagle) @ Microsoft (7 th)	294,336-core (Dell Poweredge) @ Shakti Cloud (India) (32 nd) – new
1,824,768-core (Leonardo) @CINECA in Italy (12 th)	256,960-core (Nano 4) @ NCHPC, Taiwan (33 rd)
718,848-core (ISEG2)@Nebius AI (Netherlands) (15 th)	237,280-core (Njoerd) @ Northern Data Group in the UK (35 th)
663,040-core (MareNostrum 5 ACC),@BSC in Spain (16 th)	285,696-core (Cassava AI Factory) @ Africa Data Centres (South Africa) (36 th) – new
747,520-core (Maximus-02) @ Core42 (USA) (18 th) – new	303,104-core (Harpia) @ Petróleo Brasileiro (Brazil) (37 th) – new
479,232-core (ABCI 3.0) @ AIST (Japan) (19 th) – new	315,120-core (ABCI-Q) @ AIST in Japan (38 th)
642,072-core (NIPA()-CL1) @ NHN Cloud (South Korea) (20 th) – new	36,720-core (NIPA()) @ Niagara Computers (South Korea) (39 th) – new
662,256-core (CHIE-4) @ SoftBank Corp., Japan (21 st)	320,280-core (NIPA()-CL2) @ NHN Cloud (South Korea) (40 th) – new
479,232-core (ABCI 3.0),@AIST in Japan (16 th)	223,088-core (Gefion) @ Danish Centre for AI Innovation in Denmark (43 rd)
485,888-core (Eos NVIDIA DGX SuperPOD),@NVIDIA (23 rd)	555,520-core (Selene) @ NVIDIA (45 th)
349,440-core (SSC-24)@Samsung Electronics in Korea (25 th)	185,712-core (SuperPOD) @ Core42 (UAE) (46 th) – new
297,840-core (CHIE-3),@Softbank in Japan (28 th)	and many more!

Ethernet-based Scientific Computing Installations (June 2026)

- 165 Ethernet-based (1G, 10G, 25G, 50G, 100G, 200G, 400G, 800G) compute systems with ranking in the June 2026 Top500 list
 - 11,039,616-core (El Capitan) using Slingshot-11 at LLNL (2nd)
 - 8,699,904-core (Frontier) using Slingshot-11 at ORNL (3rd)
 - 4,742,808-core (Aurora) using Slingshot-11 at ANL (4th)
 - 3,461,472-core (HPC7) using Slingshot-11 at Eni S.p.A, Italy (6th) – new
 - 3,143,520-core (HPC6) using Slingshot-11 at Eni S.p.A., Italy (7th)
 - 2,121,600-core (Alps) using Slingshot-11 at CSCS, Switzerland (10th)
 - 2,752,704-core (LUMI) using Slingshot-11 at EuroHPC, Finland (11th)
 - 1,028,160-core (Isambard-AI phase 2) using Slingshot-11 at Univ. Bristol (13th)
 - 1,161,216-core (Tuolumne) using Slingshot-11 at LLNL (14th)
 - 822,528-core (Discovery 6) using Slingshot-11 at ExxonMobil (17th)
 - 576,464-core (Shaheen III-GPU) using Slingshot 11 at KAUST, Saudi Arabia (22nd)
 - 976,896-core (MAXIMUS-384) using RoCE at Maximus, USA (24th)
 - 481,440-core (Venado) using Slingshot-11 at LANL (26th)
 - 548,5332-core (CEA-HE) using BXI-v2 at CEA in France (28th)
 - 411,264-core (DAEDALUS) using Slingshot-11 at EuroHPC in Greece (31st) – new
 - 888,832-core (Perlmutter) using Slingshot-11 at NERSC (34th)
 - 383,040-core (El Dorado) using Slingshot-11 at SNL (41st)
 - 310,080-core (Arrhenius GPU) using Slingshot-11 @ EuroHPC in Sweden (42nd) – new
 - 239,616-core (ASPIRE 2B – GPU Partition) using Slingshot-11 @ National Supercomputing Center Singapore (44th) – new
 - 483,840-core (AI-03) using Broadcom NetXtreme-E at Core42 (UAE) (47th)
 - 147,456-core (EYAS) using 400g Ethernet @ TAHAKOM (Saudi Arabia) (51st) – new
 - and many more!

Trends in Commodity Interconnects since 2015



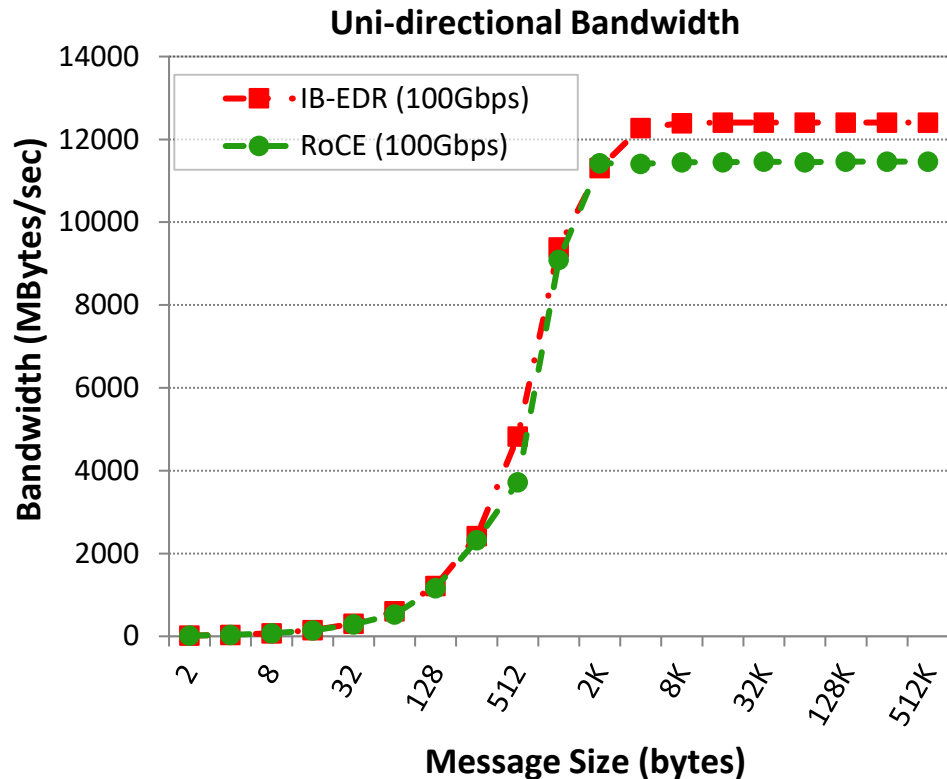
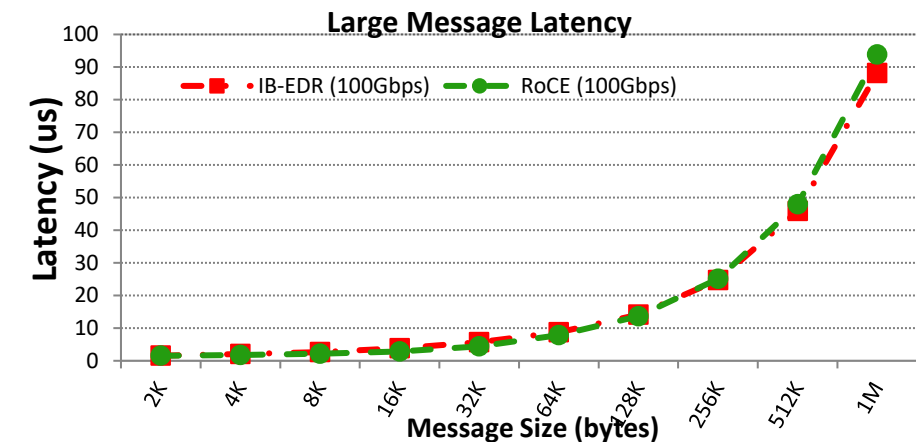
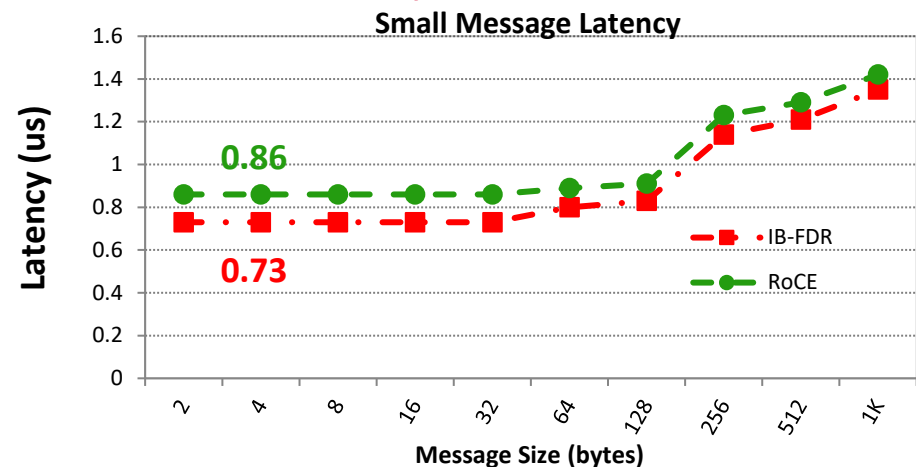
Courtesy: TOP500.org

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- **Low-level Performance**
- Message Passing Interface (MPI)

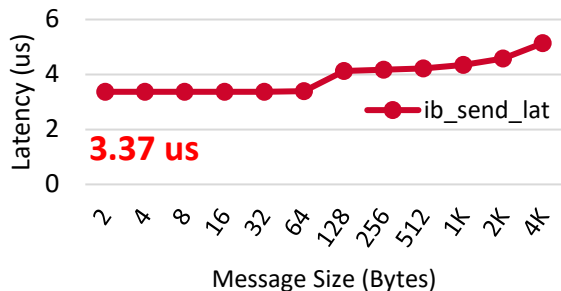
Low-level Latency and Uni-directional Bandwidth Measurements (IB-EDR v RoCE)



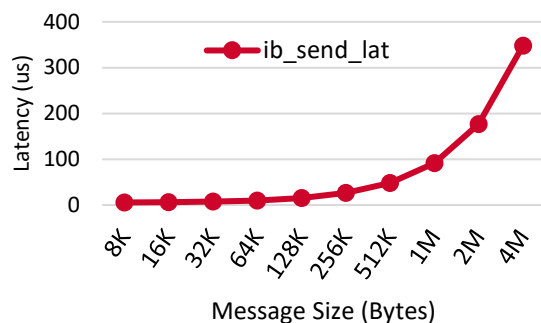
ConnectX-4 EDR (100 Gbps): 3.1 GHz Deca-core (Haswell) Intel Back-to-back
ConnectX-4 EN (100 Gbps): 3.1 GHz Deca-core (Haswell) Intel Back-to-back

Verbs Level Evaluation of Broadcom RoCE adapters

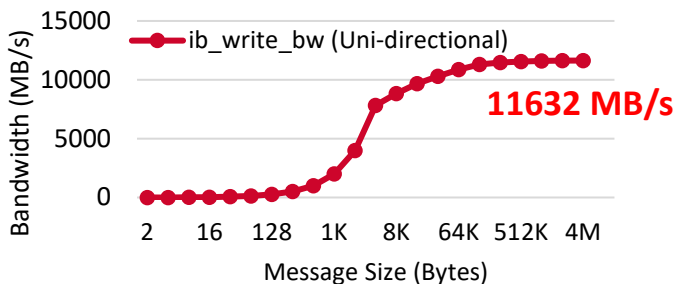
Small message Latency



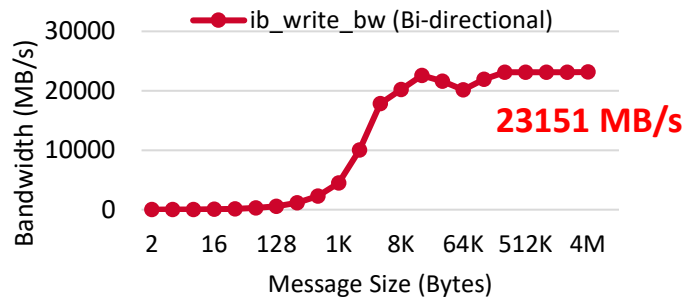
Medium/Large message Latency



Uni-directional Bandwidth



Bi-Directional Bandwidth



Broadcom NetXtreme RoCE HCA (100 Gbps): 2 GHz AMD EPYC 7662 64-Core Processor

- Low-level Performance
- **Message Passing Interface (MPI)**

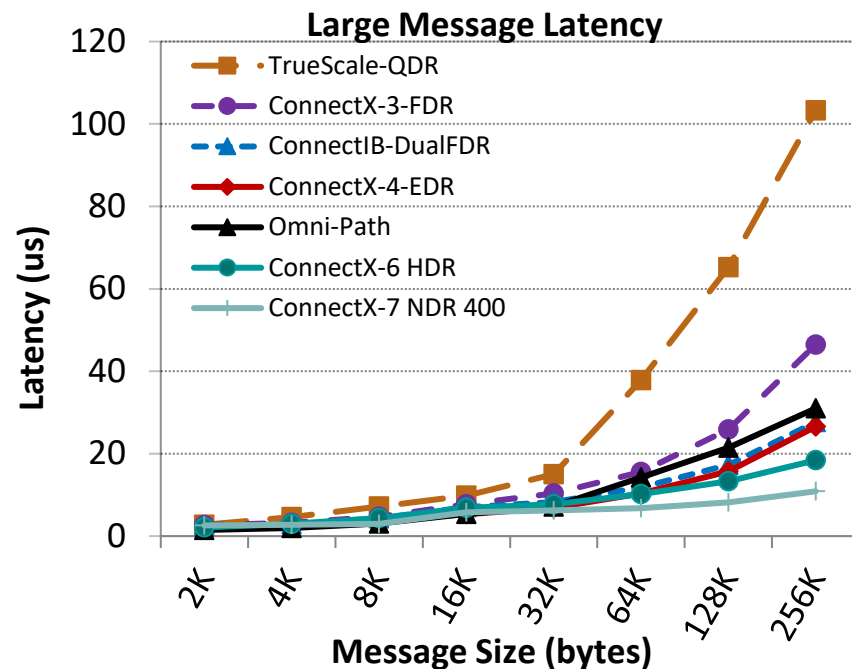
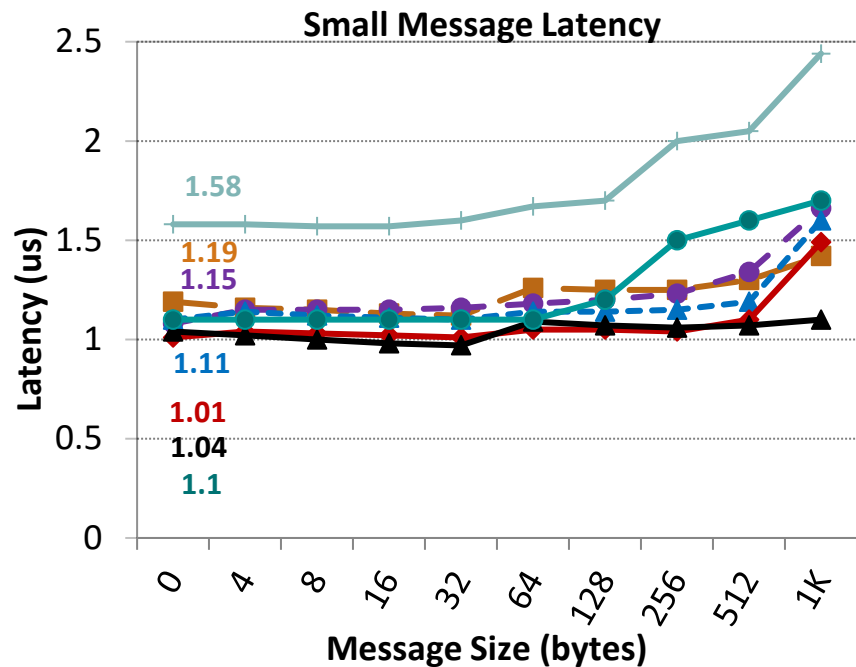
Overview of the MVAPICH Project

- High Performance open-source MPI Library
- Support for multiple interconnects
 - InfiniBand, Omni-Path, Ethernet/iWARP, RDMA over Converged Ethernet (RoCE), AWS EFA, OPX, Broadcom RoCE, Intel Ethernet, Rockport Networks, Slingshot 10/11
- Support for multiple platforms
 - x86, OpenPOWER, ARM, Xeon-Phi, GPGPUs (NVIDIA and AMD)
- Started in 2001, first open-source version demonstrated at SC '02
- Supports the latest MPI-5.0 standard
- <http://mvapich.cse.ohio-state.edu>
- Additional optimized versions for different systems/environments:
 - MVAPICH-Plus (Unification of MVAPICH2-X and MVAPICH2-GDR), since 2023
 - MVAPICH2-X (Advanced MPI + PGAS), since 2011
 - MVAPICH2-GDR with support for NVIDIA (since 2014) and AMD (since 2020) GPUs
 - MVAPICH2-MIC with support for Intel Xeon-Phi, since 2014
 - MVAPICH2-Virt with virtualization support, since 2015
 - MVAPICH2-EA with support for Energy-Awareness, since 2015
 - MVAPICH2-Azure for Azure HPC IB instances, since 2019
 - MVAPICH2-X-AWS for AWS HPC+EFA instances, since 2019
- Tools:
 - OSU MPI Micro-Benchmarks (OMB), since 2003
 - OSU InfiniBand Network Analysis and Monitoring (INAM), since 2015



- Used by more than **3,500 organizations in 94 countries** (listed under the Users Tab of the MVAPICH page)
- More than **2.09 Million downloads from the OSU site directly**
- Empowering many TOP500 clusters (Jun '26 ranking)
 - 27th, 10,649,600-core (Sunway TaihuLight) at NSC, Wuxi, China
 - 90th, 448,448 cores (Frontera) at TACC
 - 298th, 42,336 cores (Cosmos) at SDSC
- Available with software stacks of many vendors and Linux Distros (RedHat, SuSE, OpenHPC, and Spack)
- Partner in the 90th ranked TACC Frontera system
- Empowering Top500 systems for more than 20+ years

One-way Latency: MPI over IB with MVAPICH



TrueScale-QDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB switch

Omni-Path - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with Omni-Path switch

ConnectX-3-FDR - 2.8 GHz 10-core (IvyBridge) Intel PCI Gen3 with IB switch

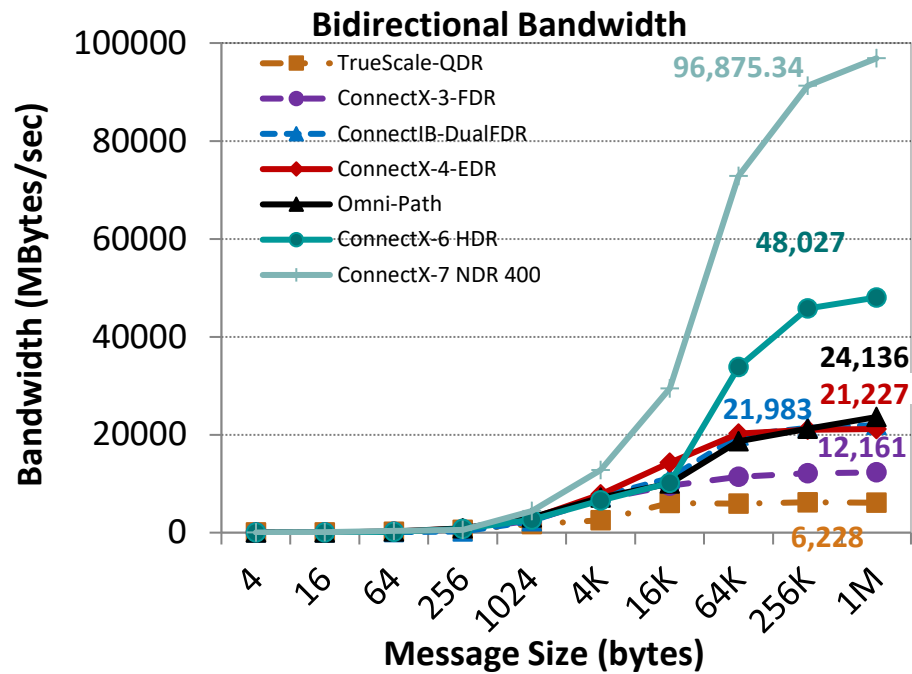
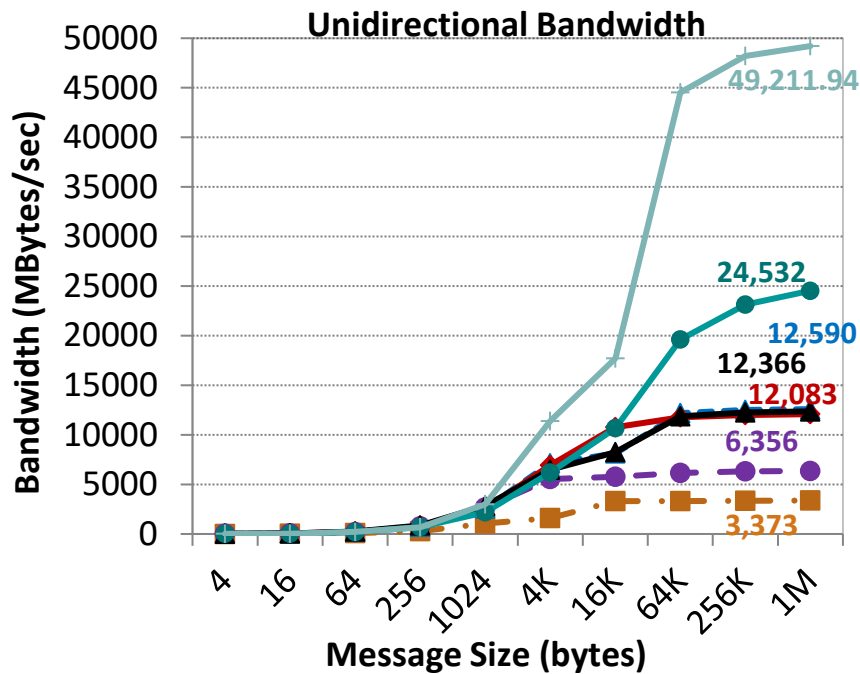
ConnectX-6-HDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB Switch

ConnectIB-Dual FDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB switch

ConnectX-7-NDR - 3.4 GHz 72-core (Grace) ARM PCIe Gen5 with IB Switch

ConnectX-4-EDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB Switch

Bandwidth: MPI over IB with MVAPICH



TrueScale-QDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB switch

Omni-Path - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with Omni-Path switch

ConnectX-3-FDR - 2.8 GHz 10-core (IvyBridge) Intel PCI Gen3 with IB switch

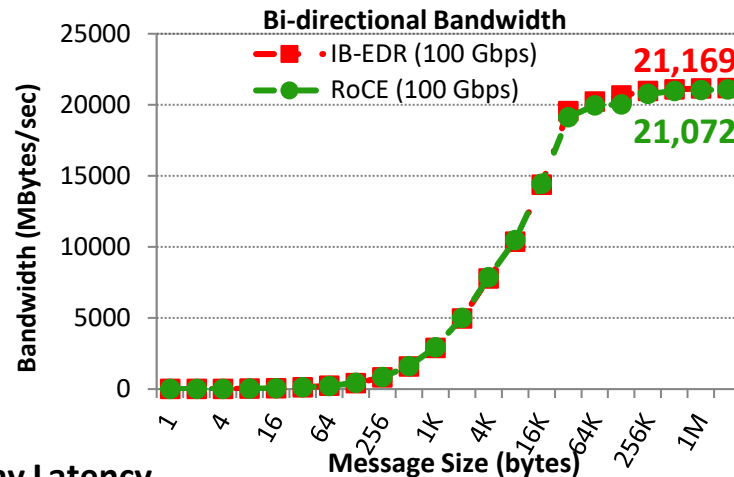
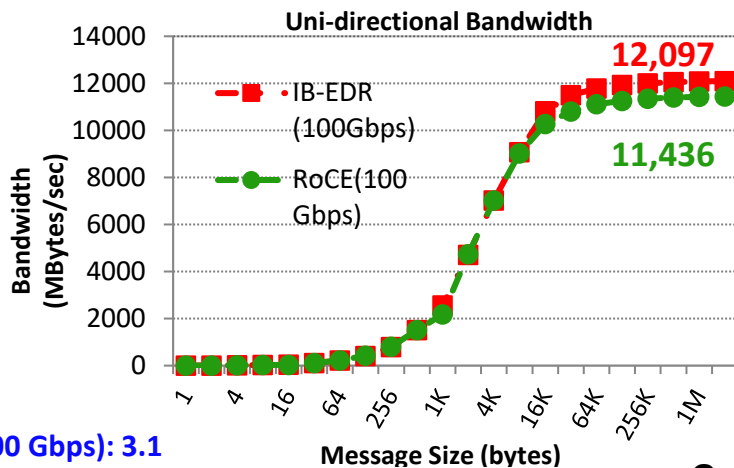
ConnectX-6-HDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB Switch

ConnectIB-Dual FDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB switch

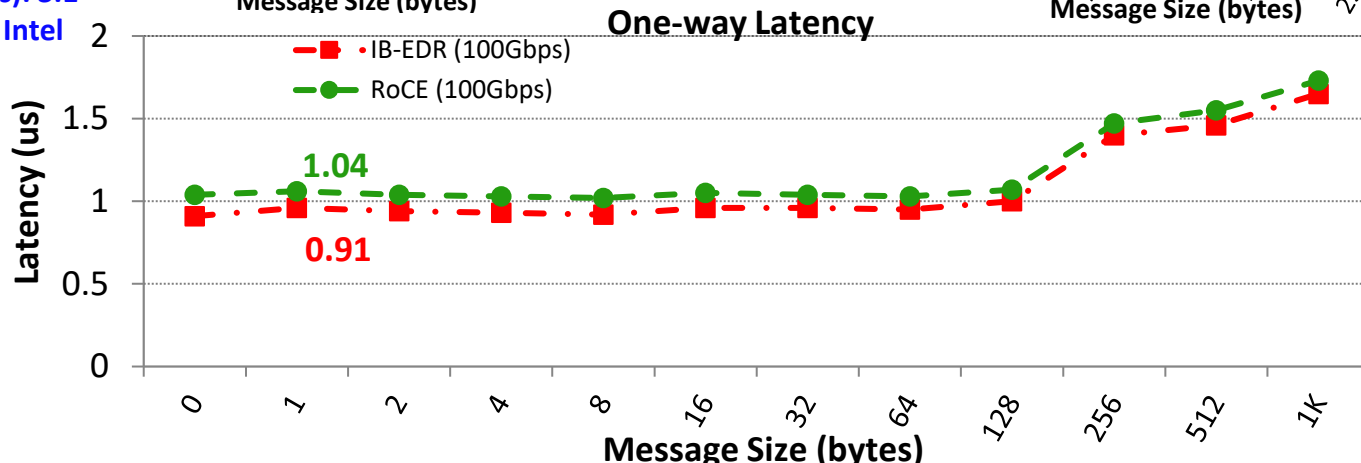
ConnectX-7-NDR - 3.4 GHz 72-core (Grace) ARM PCIe Gen5 with IB Switch

ConnectX-4-EDR - 3.1 GHz 10-core (Haswell) Intel PCI Gen3 with IB Switch

Convergent Technologies: MPI Latency and Uni-/Bi-directional BW

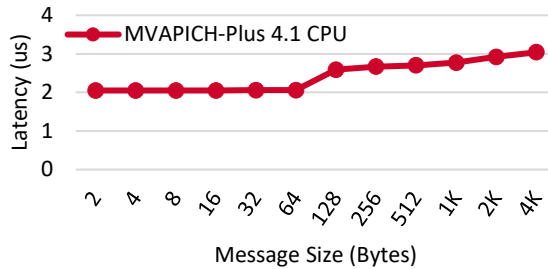


ConnectX-4 EDR (100 Gbps): 3.1
GHz Deca-core (Haswell) Intel
Back-to-back

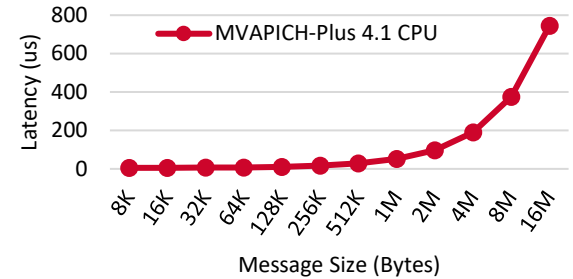


MPI Level Latency/Bandwidth on Slingshot 11

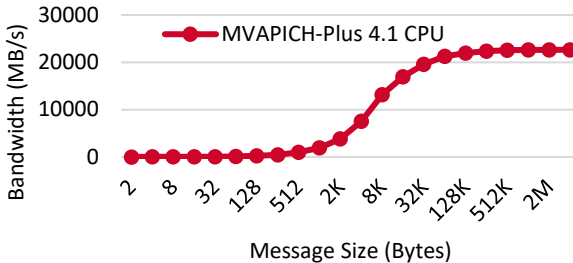
*Small message
Latency*



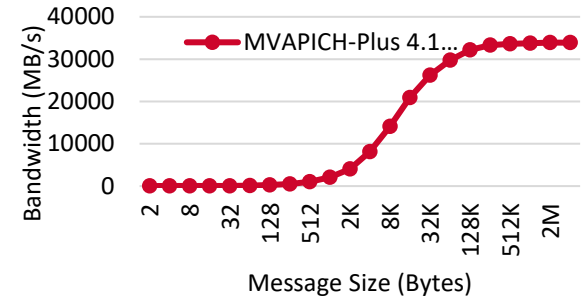
*Medium/Large
message Latency*



*Uni-directional
Bandwidth*



*Bi-Directional
Bandwidth*

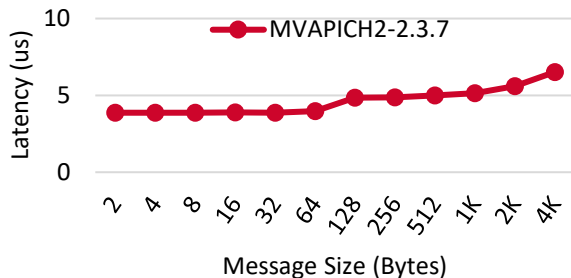


- **2us** inter-node point-to-point latency for small messages
- **22,660 MB/s** uni-directional peak bandwidth @ 4 MB
- **33,885 MB/s** bi-directional peak bandwidth @ 4MB

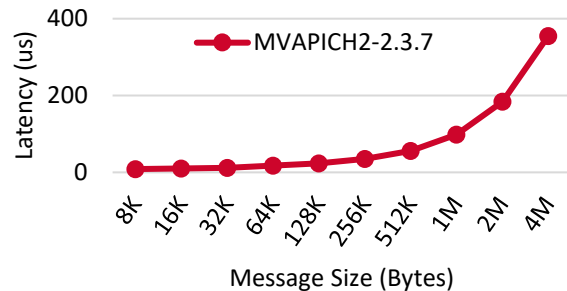
Interconnect : Cray HPE Slingshot 11, Library : MVAPICH-Plus 4.1, CPU : Frontier Supercomputer (ORNL)

MPI Level Latency/Bandwidth on Broadcom RoCE

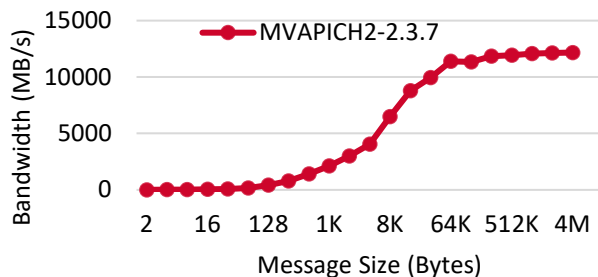
*Small message
Latency*



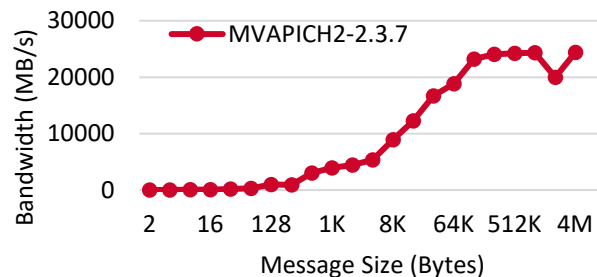
*Medium/Large
message Latency*



*Uni-directional
Bandwidth*



*Bi-Directional
Bandwidth*



- 3.88us inter-node point-to-point latency for small messages
- **12,171 MB/s** uni-directional peak bandwidth
- **24,394 MB/s** bi-directional peak bandwidth

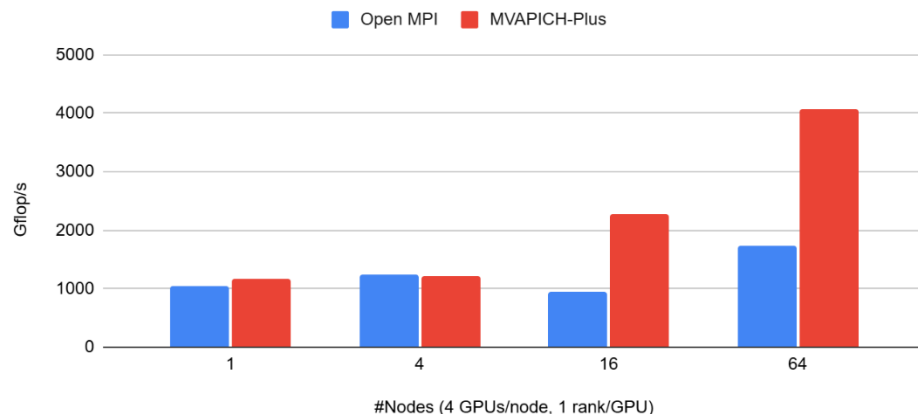
Interconnect : RoCE 100GbE with Broadcom NetXtreme Thor, Library : MVAPICH2 3.0, CPU : 2 GHz AMD EPYC 7662 64-Core Processor

Applications-Level Performance Evaluation

heFFTe on MareNostrum 5 (3D-FFT, N=1024)

- ***cuFFT backend, complex F64, Alltoallv w/ no-reordering, pencils used, GPU-Aware***
- Performance improvement of the overall execution time
 - Up to 138% improvement on runtime when running against 64 Nodes and a GPU-Aware Open MPI

heFFTe Performance on MareNostrum 5 (3D-FFT, N=1024)
(cuFFT backend, complex FP64, a2av, no-reorder, pencils, gpu-aware)



Interconnect : 4 x InfiniBand NDR-200

Library : MVAPICH-Plus 4.0rc

CPU: 2x Intel SPR 8460Y+ equipped with 4x H100 GPUs

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Getting Set-up for the Hands-on Exercise 3 (MPI-Library Tests)

- You will run the experiments on the OSU RI2 cluster

- Please use the account name and password from <http://go.osu.edu/ibtutorial>**

- Open your favorite Terminal

\$ ssh ri2tut01@ri2.cse.ohio-state.edu

Enter Password:

- Make sure to `cd` to your home directory before cloning the repository

- Clone from <https://github.com/OSU-Nowlab/HPN-Tutorial.git>
- Verify that the following files are present:

```
[michalowicz.2@ri2 HPN-Tutorial]$ ls -l
total 56
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 481 Jul 24 18:36 README.TXT
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 1801 Jul 27 13:02 run_dpu-rome.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 1379 Jul 27 13:01 run_dpu-thor.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 624 Jul 24 18:36 run_fabtest_bw.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 640 Jul 24 18:36 run_fabtest_lat.sh
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 1356 Jul 27 13:02 run_gpu.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 673 Jul 24 18:36 run_inline_ib.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 668 Jul 24 18:36 run_inline_mpi.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 683 Jul 24 18:36 run_mtu.sh
-rw-rw-r-- 1 michalowicz.2 michalowicz.2 433 Jul 24 18:36 run_omb_ofi.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 444 Jul 24 18:36 run_omb.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 498 Jul 24 18:36 run_perftest_bw.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 501 Jul 24 18:36 run_perftest_lat.sh
-rwxrwxr-x 1 michalowicz.2 michalowicz.2 866 Jul 24 18:36 run_rc_ud.sh
```

Benchmarking MPI Performance

```
$ srun -N2 --reservation=ibtutorial run_omb.sh
```

```
+ /opt/mvapich2/mvapich2-2.3.7/bin/mpirun_rsh -np 2 gpu05 gpu06 \  
MV2_HOMOGENEOUS_CLUSTER=1 MV2_IBA_HCA=mlx5_0 \  
/opt/mvapich2/mvapich2-2.3.7/libexec/osu-micro-benchmarks/mpi/pt2pt/osu_latency  
# OSU MPI Latency Test v5.9  
# Size          Latency (us)  
0                1.21  
1                1.23  
2                1.21  
4                1.19  
8                1.17  
16               1.19  
32               1.18  
64               1.20  
128              1.26
```

Benchmarking MPI Performance

#Bytes	Latency (us)	Bandwidth (MB/s)	Bi-Bandwidth(MB/s)
1	1.21	5.43	4
2	1.23	10.69	8.17
4	1.21	21.08	16.75
8	1.19	41.79	34.59
16	1.17	82.29	71.42
32	1.19	163.06	162.32
64	1.18	353.42	518.72
128	1.2	655.47	665.11
256	1.26	1333.58	1875.53
512	1.66	2291.8	3225.89
1024	1.73	3569.84	4724.21
2048	1.88	5250.27	6284.36
4096	2.22	6806.01	7664.77
8192	2.93	9669.86	10540.81
16384	4.37	7736.91	11650.45

Latency is close to IB Latency

MPI Bandwidth for large messages is close to Line Rate

MPI Collective Performance with RC vs. UD

```
$ srun -N4 --reservation=ibtutorial run_rc_ud.sh
```

Running MPI_Alltoallv with 112 processes with RC

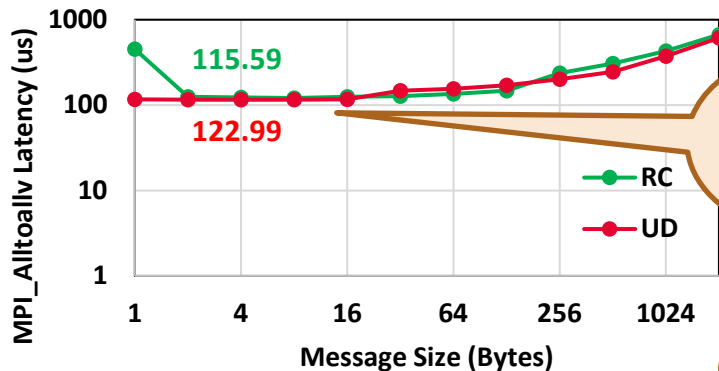
```
+ /opt/mvapich2/mvapich2-2.3.7/bin/mpirun_rsh -np 112 -hostfile $HOME/hosts-10291
MV2_HOMOGENEOUS_CLUSTER=1 MV2_IBA_HCA=mlx5_0 MV2_USE_ONLY_UD=0
/opt/mvapich2/mvapich2-2.3.7/libexec/osu-micro-
benchmarks/mpi/collective/osu_alltoallv -m 2048
...
```

Running MPI_Alltoallv with 112 processes with UD

```
+ /opt/mvapich2/mvapich2-2.3.7/bin/mpirun_rsh -np 112 -hostfile $HOME/hosts-10291
MV2_HOMOGENEOUS_CLUSTER=1 MV2_IBA_HCA=mlx5_0 MV2_USE_ONLY_UD=1
MV2_UD_MAX_RECV_WQE=512 MV2_UD_MAX_SEND_WQE=512 /opt/mvapich2/mvapich2-
2.3.7/libexec/osu-micro-benchmarks/mpi/collective/osu_alltoallv -m 2048
...
```

MPI Collective Performance with RC vs. UD

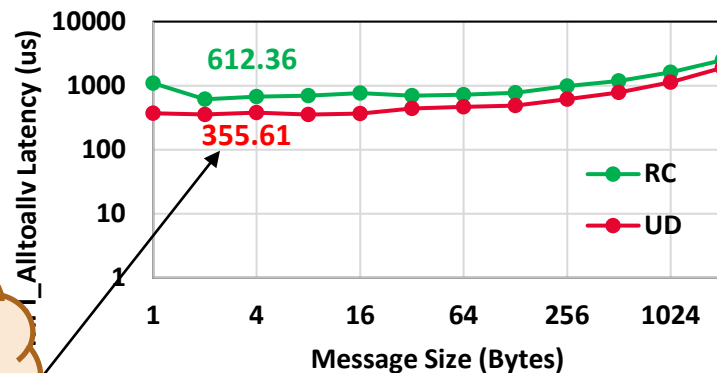
112 Processes (2 nodes, 56 ppn)



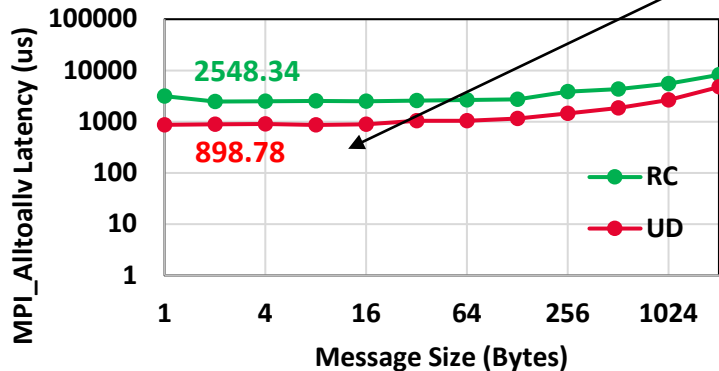
RC has less overheads at smaller job sizes

UD shows benefits as job size increases

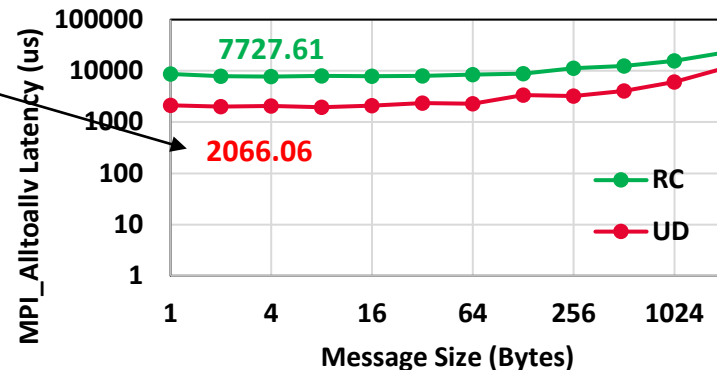
224 Processes (4 nodes, 56 ppn)



448 Processes (8 nodes, 56 ppn)



896 Processes (16 nodes, 56 ppn)



ConnectX-6-HDR100 (100 Gbps), 2x2.70 GHz 28-core Intel Cascade Lake with IB (HDR) switches

Presentation Overview

- Introduction
- Why High-Performance Networking for HPC and AI?
- RDMA Networks, Communication Models, and Semantics
- Overview of Representative Scale-Out Networks (InfiniBand and HSE)
- Hands-on Exercises-1: InfiniBand
- Overview of Modern Scale-Out RDMA Networks
 - Omni-Path Interconnect Architecture
 - Cray/HPE Slingshot Interconnect Architecture
 - Ultra-Ethernet Consortium (UEC)
- Overview of Scale-Up Networks
 - NVIDIA NVLink/NVSwitch Interconnect Architecture
 - AMD Infinity Fabric/XGMI Interconnect Architecture
 - Compute Express Link (CXL)
 - Ultra-Accelerator Link Consortium (UALink)
 - Broadcom Scale-Up Ethernet and OCP's Ethernet Scale-Up Networks
- Overview of Software Stacks for Commodity High-Performance Networks
 - OpenFabrics
 - UCX and Libfabrics
- GPU-Aware Communication
 - GPU-Direct RDMA (GDR), IB GPUDirect Async (IBGDA)
 - Hands-On Exercises 2: Use of GDR Technology in MPI
- Overview of Emerging Smart Network Interfaces
 - Collectives with NVIDIA SHARP
 - NVIDIA BlueField DPUs, AMD Pensando SmartNICs, and Intel IPUs
- High-Performance Network Deployments for AI Workloads
 - Cerebras, Maia, and MTIA
- Network trends in the Top500
- Sample and Performance Numbers
- Exercise 3: MPI Technologies
- **Conclusions and Final Q&A**

Concluding Remarks

- Presented smart networking architectures & trends in Clusters
- Presented background and details of various smart network for HPC and AI
 - Highlighted the main features of high-performance networks
 - Gave an overview of high-performance network hardware/software ecosystem
 - Discussed sample performance numbers in designing various high-end systems
- Smart networking architectures are leading to a new generation of networked computing systems, opening many research issues needing novel solutions
- Will see many more innovations in the coming years for exascale/zetascale systems

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Equipment Support by



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Current Students (Under/Graduate)

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1. Join our half-day tutorials at IEEE Hot Interconnects 2026!

- **Tutorial: Principles and Practice of Scalable and Distributed AI Training and Inference**
 - Friday, August 21, 13:30 - 16:30 Pacific Time (PT)
- **Tutorial: High Performance and Smart Networking Technologies for HPC and AI**
 - Friday, August 21, 09:00 - 12:00 Pacific Time (PT)

Hot Interconnects Registration (free) <https://hoti.org/register.html>

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 - Nov 15-20 (Exact date/time TBD)

3. CARLA Conference in Córdoba, Argentina!

- **Tutorial: Principles and Practice of Scalable and Distributed AI Training and Inference**
 - September 21, 2026, 13:30 - 17:00 Eastern Time (ET) – 14:30 – 18:00 Córdoba Time
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 - Monday, September 21, 2026, 8:00 – 11:00 Eastern Time (ET) – 9:00 – 12:00 Córdoba Time

14th Annual MVAPICH User Group (MUG) Conference

<http://mug.mvapich.cse.ohio-state.edu/>

- August 17-19, 2025, Columbus, Ohio, USA

- **Keynote Speakers:**

- Yutong Lu, Sun Yat-Sen University
- Dan Stanzione, Texas Advanced Computing Center (TACC)

- **Invited Speakers:**

- Ahmad Abdelfattah and Natalie Beams, University of Tennessee, Knoxville
- Sameh Abdulah, King Abdullah University of Science and Technology (KAUST), Saudi Arabia
- HooYoung Ahn, ETRI, South Korea
- Jeff Hammond, NVIDIA
- Jithin Jose, Microsoft
- Niranjana Vaidya and Hemal Shah, Broadcom
- Solyman Ashrafi, Fujitsu/1Finity
- Soham Ghosh and Kyle Schaefer, X-ScaleSolutions
- Charles Archer and Brian Hwang, Cornelis Networks
- Ankur Sheth, Dan Mihailescu, and Eric Yu, Keysight; Mihai-Gabriel Constantin, UPB Romania

- **Invited Speakers (Cont'd):**

- Vipin Chaudhary, Case Western Reserve University
- Richard Graham, NVIDIA
- Rakesh Kumar Yadav, CDAC, India
- Sameer Shende, ParaTools and University of Oregon
- Mahidhar Tatineni, San Diego Supercomputer Center (SDSC)
- Xiaoyi Lu, University of Florida
- Sangamesh Kodge, Sai Rahul Chalamalasetti, and David Karlov, D-Matrix
- Hari Subramoni and Brijesh Nanda, OSU

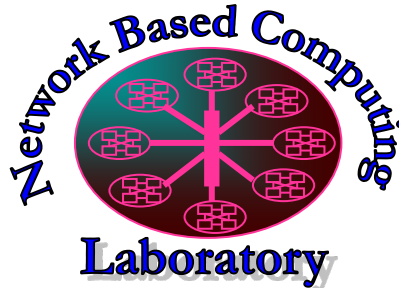
- **Tutorials and Demos:**

- 1Finity
- Keysight
- D-matrix
- OSU MVAPICH
- OSU HiDL
- ParaTools
- X-ScaleSolutions

**Slides and Videos of the
Talks are being uploaded!**

Thank You!

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